QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

trezor-firmware-en @ 72489948b3c4 · SatoshiLabs (Trezor) — public repository (passive review)

Organization	QSI
Report date	2026-11-04 19:34 UTC
Report version	1
Audit run	80f0642e-51ac-4e04-8c1c-fa6fcc6b2df6 · Document audit · 2026-11-04 19:34 UTC
Analyzer engine	qsi-engine 1.3.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`90dd7412a16c9c32e0ab86bd7ea84f18277cb6f27e0cc88c4eafa5a87e5841cc`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

11	0	0	11	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	0
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	11
Improvement recommendation	0
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-12887	Medium	Transaction signing for 11 chain(s) relies on elliptic-curve signatures confirmed by the analyst	56.8
QSI-2026-12877	Medium	Quantum-vulnerable Ed25519: firmware and boot verification (root of trust) (2 file(s)) confirmed by the analyst	43.4
QSI-2026-12879	Medium	Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (3 file(s)) confirmed by the analyst	41.8
QSI-2026-12880	Medium	Quantum-vulnerable ECDSA: transaction and message signing (4 file(s)) confirmed by the analyst	41.8
QSI-2026-12883	Medium	Quantum-vulnerable Ed25519: transaction and message signing (3 file(s)) confirmed by the analyst	41.8

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Analyst review: 11 confirmed, 0 not applicable, 0 dismissed as false positives.

Post-quantum maturity: Level 3 of 4 — hybrid protection in a critical path. Next step: Cover the remaining classical critical roles and plan when the classical half can be dropped.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/trezor/trezor-firmware @ 72489948b3c44a37598de9da658be28e4fabb4ae — evidence links open the lines at this commit.

Supplied inputs

411 files, 2378051 bytes in total, 0 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.3.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	374	common/protob/check.py: 49 lines scanned with 16 rule patterns and 145 registry patterns; common/protob/graph.py: 76 lines scanned with 16 rule patterns and 145 registry patterns; core/embed/projects/bootloader/bootui.c: 139 lines scanned with 16 rule patterns and 145 registry patterns
chain_map	checked	13	common/protob/messages-benchmark.proto: Bitcoin; common/protob/messages-bitcoin.proto: Bitcoin; common/protob/messages-cardano.proto: Cardano
dependencies	checked	2	core/embed/projects/bootloader/Cargo.toml: 9 dependencies parsed (0 pinned); core/embed/sec/Cargo.toml: 6 dependencies parsed (0 pinned)
certificates	error (not checked)	4	core/embed/sec/optiga/unix/certs/T2B1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError); core/embed/sec/optiga/unix/certs/T3B1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError); core/embed/sec/optiga/unix/certs/T3T1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
certificates_and_keys	checked	5	core/embed/sec/optiga/unix/certs/T2B1.der: certificates/keys parsed; core/embed/sec/optiga/unix/certs/T3B1.der: certificates/keys parsed; core/embed/sec/optiga/unix/certs/T3T1.der: certificates/keys parsed
entropy	checked	1	random-number generation: 373 code files read with 722 macro definitions from the tree; 0 ineffective guard(s), 0 weak random-bytes provider(s), 0 weak key-generation site(s)
osv_advisories	not checked	1	no exactly pinned versions to query

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 395 (input files: C)

Not checked

- certificates: core/embed/sec/optiga/unix/certs/T2B1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- certificates: core/embed/sec/optiga/unix/certs/T3B1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- certificates: core/embed/sec/optiga/unix/certs/T3T1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- certificates: core/embed/sec/optiga/unix/certs/T3W1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- osv_advisories: no exactly pinned versions to query

Architecture overview

Static review of the source tree trezor-firmware-en at 72489948b3c44a37598de9da658be28e4fabb4ae. 411 files analysed; of 887 files, 441 are test, fuzzing, benchmark or example code and were left out, 35 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. The review was restricted to 4 path(s) of the repository: common/protob, core/embed/projects/bootloader, core/embed/sec, crypto; 24659 files outside them were not read. Files no analyzer reads include: common/protob/.gitignore, common/protob/Makefile, common/protob/pb2py, common/protob/protocol.md, core/embed/projects/bootloader/.changelog.d/.gitignore. No requests were sent to the owner's systems.

Overall security posture

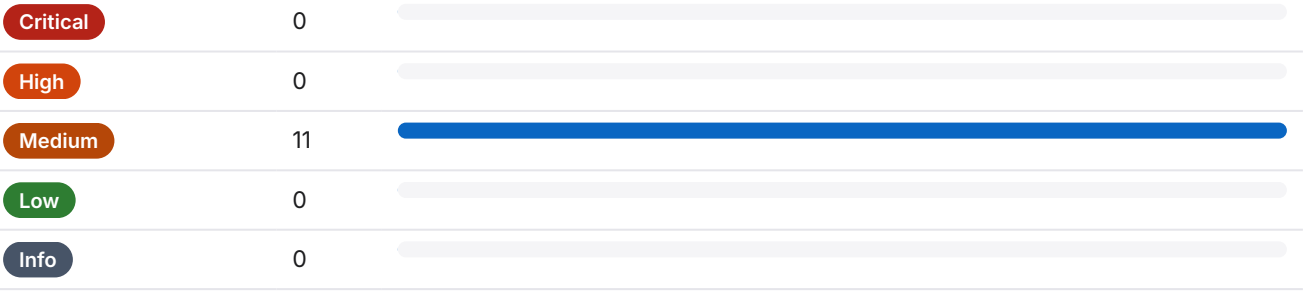
100
Indicative posture index (0–100)

66.1%
Quantum-vulnerable share of inventory

112
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-12887	Transaction signing for 11 chain(s) relies on elliptic-curve signatures	Long-term post-quantum risk	Medium	High
QSI-2026-12877	Quantum-vulnerable Ed25519: firmware and boot verification (root of trust) (2 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12879	Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (3 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12880	Quantum-vulnerable ECDSA: transaction and message signing (4 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12883	Quantum-vulnerable Ed25519: transaction and message signing (3 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12886	Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12878	Quantum-vulnerable ECDSA: authentication and identity (certificates, tokens, device attestation) (2 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12881	Quantum-vulnerable ECDH: role not determined (2 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12882	Quantum-vulnerable X25519: role not determined (3 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12884	Quantum-vulnerable ECDSA: role not determined (8 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-12885	Quantum-vulnerable Ed25519: role not determined (4 file(s))	Long-term post-quantum risk	Medium	Medium

Detailed findings

QSI-2026-12887 · CHAINMAP-QV

Medium

Transaction signing for 11 chain(s) relies on elliptic-curve signatures

Long-term post-quantum risk | Confidence: High

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** True by construction and the most important row in the report for planning. The protobuf message definitions in common/protob cover 11 chains whose signature schemes are fixed by consensus, not by the wallet: Bitcoin, Cardano, EOS, Ethereum and EVM, Monero, NEM, Solana, Stellar, Tezos, Tron, XRP. The wallet cannot migrate any of them unilaterally, so the deliverable here is a readiness map, not a code change — which is exactly why this belongs in an inventory rather than a vulnerability list.

The wallet's message definitions cover Bitcoin, Cardano, EOS, Ethereum and EVM chains, Monero, NEM, Solana, Stellar, Tezos, Tron, XRP Ledger. These chains authorize transactions with ECDSA, Ed25519, Schnorr (BIP-340), which Shor's algorithm breaks on a cryptographically relevant quantum computer. A wallet cannot move a chain to post-quantum signatures on its own: the chain has to adopt a scheme first.

Affected asset	chain-map
Data source	common/protob/messages-benchmark.proto
Detection method	Message definitions (.proto) matched to QSI's table of chain signature schemes.
Analyzer	chain_map 1.3.0 · 2026-11-04 19:34 UTC
Priority score	56.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Once a CRQC exists, an attacker who sees an account's public key could derive the private key and move funds; how soon keys are exposed depends on each chain (address reuse, account model).

REPRODUCTION STEPS

```
Open the .proto files named in the evidence.
```

EVIDENCE

file_excerpt · common/protob/messages-benchmark.proto · line 2 · SHA-256 d58a0f922546e558...

```
package hw.trezor.messages.bitcoin;
```

file_excerpt · common/protob/messages-bitcoin.proto · line 2 · SHA-256 d58a0f922546e558...

```
package hw.trezor.messages.bitcoin;
```

file_excerpt · common/protob/messages-cardano.proto · line 2 · SHA-256 790478e3c9df02a9...

```
package hw.trezor.messages.cardano;
```

file_excerpt · common/protob/messages-eos.proto · line 2 · SHA-256 1bc71146f1570006...

```
package hw.trezor.messages.eos;
```

file_excerpt · common/protob/messages-ethereum-eip712.proto · line 2 · SHA-256 1e20c49087fd2d82...

```
package hw.trezor.messages.ethereum_eip712;
```

file_excerpt · common/protob/messages-ethereum.proto · line 2 · SHA-256 2c715738dc6f29a4...

```
package hw.trezor.messages.ethereum;
```

file_excerpt · common/protob/messages-monero.proto · line 2 · SHA-256 2a60fdbae9c4da47...

```
package hw.trezor.messages.monero;
```

file_excerpt · common/protob/messages-nem.proto · line 2 · SHA-256 ae6d3c9d0ee3ae3e...

```
package hw.trezor.messages.nem;
```

file_excerpt · common/protob/messages-ripple.proto · line 2 · SHA-256 fc5096c71f30511d...

```
package hw.trezor.messages.ripple;
```

file_excerpt · common/protob/messages-solana.proto · line 2 · SHA-256 bfc45618be53d3fa...

```
package hw.trezor.messages.solana;
```

file_excerpt · common/protob/messages-stellar.proto · line 2 · SHA-256 fad5a4b7a72a7677...

```
package hw.trezor.messages.stellar;
```

file_excerpt · common/protob/messages-tezos.proto · line 2 · SHA-256 967d28a5f166e7d4...

```
package hw.trezor.messages.tezos;
```

file_excerpt · common/protob/messages-tron.proto · line 2 · SHA-256 19b3678a04d39ff5...

```
package hw.trezor.messages.tron;
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Chains are recognized from the names of the message definitions (.proto files and packages); the schemes are the chains' documented account signatures, not an analysis of the wallet's signing code.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them. Keep the signing layer crypto-agile per chain (algorithm identifiers in messages and storage, room for larger keys and signatures) and track each chain's post-quantum proposals, so that a new scheme can be added when a chain activates it.

RETEST PROCEDURE

Re-run when a chain activates a post-quantum signature scheme or the wallet adds one.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-12877 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: firmware and boot verification (root of trust) (2 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry a role the engine determined and review agrees with, each checked against the cited file. Ed25519 at core/embed/sec/image/boot_header.c:81 and image.c is firmware and boot verification, with the public keys fixed in the bootloader. **It must not be called unmigrated, because this vendor has already started**: SLH-DSA appears in that same boot_header.c at line 30, there is a whole ML-DSA-44 module at core/embed/sec/mldsa44/, MCU attestation uses ML-DSA (core/embed/sec/mcu_attestation/), and the tree ships an X.509 certificate with an ML-DSA-44 public key (core/embed/sec/secret/unix/certs/T3W1.der). The inventory carries those seven rows as post-quantum. So the correct reading of this repository is a root of trust **mid-transition**, carrying both schemes, and the Ed25519 rows are what remains of the classical half rather than evidence that nothing has been done. Trezor is the most advanced repository in this portfolio on post-quantum readiness, and any report that presents it otherwise is wrong on a point its own engineers would see immediately. X25519 at core/embed/sec/tropic/tropic.c:383 plus crypto/noise_kk1.c and noise_xxpsk3.c is channel key exchange with the Tropic01 secure element over Noise — recordable now, decryptable later, and the standard answer is a hybrid exchange. ECDSA at crypto/bip32.c:40, crypto/secp256k1.c and crypto/zkp_bip340.c, and Schnorr (BIP-340) at crypto/zkp_bip340.c:139, are transaction signing. Ed25519 at crypto/bip32.c:477, crypto/nem.c and nem.h is transaction signing on the NEM path.

4 reference(s) to Ed25519 across 2 file(s), e.g. core/embed/sec/image/boot_header.c:81. Role, judged from paths, function names and nearby code: firmware and boot verification (root of trust). These public keys are fixed in the bootloader or the device: with a cryptographically relevant quantum computer (CRQC) an attacker forges a firmware signature, and replacing the keys needs a bootloader update in the field.

Affected asset	source-code:Ed25519:root_of_trust
Data source	core/embed/sec/image/boot_header.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	43.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open core/embed/sec/image/boot_header.c:81 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · core/embed/sec/image/boot_header.c · line 81 · SHA-256 b2af0c1adee8c4fb... · File 4506855d1b81cc19...

```
 80 |     int ec_result =
 81 |         ed25519_sign_open(hash, sizeof(hash), BOARDLOADER_EC_KEYS[key_idx],
 82 |                           sig->ec_signature[sig_idx]);
```

file_excerpt · core/embed/sec/image/image.c · line 190 · SHA-256 dfcc4a69e8899744... · File 3977fe12259f229c...

```
189 |   return sectrue *
190 |          (0 == ed25519_sign_open(fingerprint, IMAGE_HASH_DIGEST_LENGTH, pub,
191 |                                  *(const ed25519_signature *)hdr->sig));
```

file_excerpt · core/embed/sec/image/image.c · line 249 · SHA-256 708bc289abcbd866... · File 3977fe12259f229c...

```
248 |   return sectrue *
249 |          (0 == ed25519_sign_open(fingerprint, IMAGE_HASH_DIGEST_LENGTH, pub,
250 |                                  *(const ed25519_signature *)hdr->sig));
```

file_excerpt · core/embed/sec/image/image.c · line 411 · SHA-256 69cfc8f4b867ff5b... · File 3977fe12259f229c...

```
410 |   return sectrue *
411 |          (0 == ed25519_sign_open(hash, IMAGE_HASH_DIGEST_LENGTH, pub,
412 |                                  *(const ed25519_signature *)vhdr->sig));
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Plan a hybrid firmware signature (the classical one plus ML-DSA or SLH-DSA, both required) and the path to ship the new keys in a bootloader update.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12879 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (3 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry a role the engine determined and review agrees with, each checked against the cited file. Ed25519 at core/embed/sec/image/boot_header.c:81 and image.c is firmware and boot verification, with the public keys fixed in the bootloader. **It must not be called unmigrated, because this vendor has already started**: SLH-DSA appears in that same boot_header.c at line 30, there is a whole ML-DSA-44 module at core/embed/sec/mldsa44/, MCU attestation

> uses ML-DSA (core/embed/sec/mcu_attestation/), and the tree ships an X.509 certificate with an ML-DSA-44 public key (core/embed/sec/secret/unix/certs/T3W1.der). The inventory carries those seven rows as post-quantum. So the correct reading of this repository is a root of trust **mid-transition**, carrying both schemes, and the Ed25519 rows are what remains of the classical half rather than evidence that nothing has been done. Trezor is the most advanced repository in this portfolio on post-quantum readiness, and any report that presents it otherwise is wrong on a point its own engineers would see immediately. X25519 at core/embed/sec/tropic/tropic.c:383 plus crypto/noise_kk1.c and noise_xxpsk3.c is channel key exchange with the Tropic01 secure element over Noise — recordable now, decryptable later, and the standard answer is a hybrid exchange. ECDSA at crypto/bip32.c:40, crypto/secp256k1.c and crypto/zkp_bip340.c, and Schnorr (BIP-340) at crypto/zkp_bip340.c:139, are transaction signing. Ed25519 at crypto/bip32.c:477, crypto/nem.c and nem.h is transaction signing on the NEM path.

14 reference(s) to X25519 across 3 file(s), e.g. core/embed/sec/tropic/tropic.c:383. Role, judged from paths, function names and nearby code: channel key exchange (TLS, Noise, handshakes). Traffic protected by this key exchange can be recorded now and decrypted later; a hybrid exchange keeps the classical guarantees and adds post-quantum ones.

Affected asset	source-code:X25519:channel
Data source	core/embed/sec/tropic/tropic.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open core/embed/sec/tropic/tropic.c:383 and inspect the quoted lines; the evidence lists further
files.
```

EVIDENCE

file_excerpt · core/embed/sec/tropic/tropic.c · line 383 · SHA-256 6a8b8eac9f12e02d… · File 06c12cd103221953…

```
382 |   curve25519_key trezor_public = {0};
383 |   curve25519_scalarmult_basepoint(trezor_public, trezor_private);
384 |
```

file_excerpt · crypto/noise_kk1.c · line 187 · SHA-256 92dfee834b4c3685… · File 2ddf35ff89fa3390…

```
186 |                 sizeof(ctx->initiator_ephemeral_private_key));
187 |   curve25519_scalarmult_basepoint(request->initiator_ephemeral_public_key,
188 |                                   ctx->initiator_ephemeral_private_key);
```

file_excerpt · crypto/noise_kk1.c · line 199 · SHA-256 561606d6dbe7a123… · File 2ddf35ff89fa3390…

```
198 |   curve25519_key responder_public_key = {0};
199 |   curve25519_scalarmult_basepoint(responder_public_key, responder_private_key);
200 |
```

file_excerpt · crypto/noise_kk1.c · line 205 · SHA-256 e17fd3f06c5cea71… · File 2ddf35ff89fa3390…

```
204 |   curve25519_key responder_ephemeral_public_key = {0};
205 |   curve25519_scalarmult_basepoint(responder_ephemeral_public_key,
206 |                                   responder_ephemeral_private_key);
```

file_excerpt · crypto/noise_xxpsk3.c · line 138 · SHA-256 9648d44dace3fff1… · File 5550a2dcad0d68de…

```
137 |               uint8_t (*output)[NOISE_XXPSK3_DHLEN]) {
138 |   curve25519_scalarmult(*output, *private_key, *public_key);
139 | }
```

file_excerpt · crypto/noise_xxpsk3.c · line 197 · SHA-256 4e128b42ab2d3d6f… · File 5550a2dcad0d68de…

```
196 |   (*private_key)[31] |= 64;
197 |   curve25519_scalarmult_basepoint(*public_key, *private_key);
198 | }
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols).

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12880 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (4 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry a role the engine determined and review agrees with, each checked against the cited file. Ed25519 at core/embed/sec/image/boot_header.c:81 and image.c is firmware and boot verification, with the public keys fixed in the bootloader. **It must not be called unmigrated, because this vendor has already started**: SLH-DSA appears in that same boot_header.c at line 30, there is a whole ML-DSA-44 module at core/embed/sec/mldsa44/, MCU attestation uses ML-DSA (core/embed/sec/mcu_attestation/), and the tree ships an X.509 certificate with an ML-DSA-44 public key (core/embed/sec/secret/unix/certs/T3W1.der). The inventory carries those seven rows as post-quantum. So the correct reading of this repository is a root

> of trust **mid-transition**, carrying both schemes, and the Ed25519 rows are what remains of the classical half rather than evidence that nothing has been done. Trezor is the most advanced repository in this portfolio on post-quantum readiness, and any report that presents it otherwise is wrong on a point its own engineers would see immediately. X25519 at core/embed/sec/tropic/tropic.c:383 plus crypto/noise_kk1.c and noise_xxpsk3.c is channel key exchange with the Tropic01 secure element over Noise — recordable now, decryptable later, and the standard answer is a hybrid exchange. ECDSA at crypto/bip32.c:40, crypto/secp256k1.c and crypto/zkp_bip340.c, and Schnorr (BIP-340) at crypto/zkp_bip340.c:139, are transaction signing. Ed25519 at crypto/bip32.c:477, crypto/nem.c and nem.h is transaction signing on the NEM path.

13 reference(s) to ECDSA across 4 file(s), e.g. crypto/bip32.c:40. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	crypto/bip32.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crypto/bip32.c:40 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crypto/bip32.c · line 40 · SHA-256 a9165bffdef5ae27… · File 2429e49ef1cc89c5…

```
  39 | #include "nist256p1.h"
  40 | #include "secp256k1.h"
  41 | #include "sha2.h"
```

file_excerpt · crypto/bip32.c · line 670 · SHA-256 81ebab5faffac9ea… · File 2429e49ef1cc89c5…

```
669 |   if (node->curve->params) {
670 |     return ecdsa_sign_digest(node->curve->params, node->private_key, digest,
671 |                              sig, pby, is_canonical);
```

file_excerpt · crypto/secp256k1.c · line 24 · SHA-256 aaefce0c635dd202… · File 0a3762230f652a87…

```
  23 |
  24 | #include "secp256k1.h"
  25 |
```

file_excerpt · crypto/secp256k1.c · line 26 · SHA-256 3fcb800f4c815d9d… · File 0a3762230f652a87…

```
25 |
26 | const ecdsa_curve secp256k1 = {
27 |     /* .prime */ {/*.val =*/{0x1ffffc2f, 0x1fffffff, 0x1fffffff, 0x1fffffff,
```

file_excerpt · crypto/secp256k1.c · line 55 · SHA-256 7e89b542010777ba... · File 0a3762230f652a87...

```
54 |     {
55 | #include "secp256k1.table"
56 |     }
```

file_excerpt · crypto/secp256k1.h · line 32 · SHA-256 5257b883e95e417a... · File ea553cc2e389005f...

```
31 |
32 | extern const ecdsa_curve secp256k1;
33 | extern const curve_info secp256k1_info;
```

file_excerpt · crypto/zkp_bip340.c · line 30 · SHA-256 ef19a8a2b914c523... · File 05e714da983b70ec...

```
29 |
30 | #include "vendor/secp256k1-zkp/include/secp256k1.h"
31 | #include "vendor/secp256k1-zkp/include/secp256k1_extrakeys.h"
```

file_excerpt · crypto/zkp_bip340.c · line 31 · SHA-256 860dd605d6fca510... · File 05e714da983b70ec...

```
30 | #include "vendor/secp256k1-zkp/include/secp256k1.h"
31 | #include "vendor/secp256k1-zkp/include/secp256k1_extrakeys.h"
32 | #include "vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h"
```

file_excerpt · crypto/zkp_bip340.c · line 32 · SHA-256 f0d785deb2e91ec8... · File 05e714da983b70ec...

```
31 | #include "vendor/secp256k1-zkp/include/secp256k1_extrakeys.h"
32 | #include "vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h"
33 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12883 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: transaction and message signing (3 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry a role the engine determined and review agrees with, each checked against the cited file. Ed25519 at core/embed/sec/image/boot_header.c:81 and image.c is firmware and boot verification, with the public keys fixed in the bootloader. **It must not be called unmigrated, because this vendor has already started**: SLH-DSA appears in that same boot_header.c at line 30, there is a whole ML-DSA-44 module at core/embed/sec/mldsa44/, MCU attestation uses ML-DSA (core/embed/sec/mcu_attestation/), and the tree ships an X.509 certificate with an ML-DSA-44 public key (core/embed/sec/secret/unix/certs/T3W1.der). The inventory carries those seven rows as post-quantum. So the correct reading of this repository is a root of trust **mid-transition**, carrying both schemes, and the Ed25519 rows are what remains of the classical half rather than evidence that nothing has been done. Trezor is the most advanced repository in this portfolio on post-quantum readiness, and any report that presents it otherwise is wrong on a point its own engineers would see immediately. X25519 at core/embed/sec/tropic/tropic.c:383 plus crypto/noise_kk1.c and noise_xxpsk3.c is channel key exchange with the Tropic01 secure element over Noise — recordable now, decryptable later, and the standard answer is a hybrid exchange. ECDSA at crypto/bip32.c:40, crypto/secp256k1.c and crypto/zkp_bip340.c, and Schnorr (BIP-340) at crypto/zkp_bip340.c:139, are transaction signing. Ed25519 at crypto/bip32.c:477, crypto/nem.c and nem.h is transaction signing on the NEM path.

10 reference(s) to Ed25519 across 3 file(s), e.g. crypto/bip32.c:477. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Ed25519:transaction_signing
Data source	crypto/bip32.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crypto/bip32.c:477 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crypto/bip32.c · line 477 · SHA-256 c93b42f00e937006… · File 2429e49ef1cc89c5…

```
476 |     if (node->curve == &ed25519_info) {
477 |       ed25519_publickey(node->private_key, node->public_key + 1);
478 |     } else if (node->curve == &ed25519_sha3_info) {
```

file_excerpt · crypto/bip32.c · line 479 · SHA-256 8ed678007d138957… · File 2429e49ef1cc89c5…

```
478 |     } else if (node->curve == &ed25519_sha3_info) {
479 |       ed25519_publickey_sha3(node->private_key, node->public_key + 1);
480 | #if USE_KECCAK
```

file_excerpt · crypto/bip32.c · line 482 · SHA-256 183902ffc93a3093... · File 2429e49ef1cc89c5...

```
481 |     } else if (node->curve == &ed25519_keccak_info) {
482 |       ed25519_publickey_keccak(node->private_key, node->public_key + 1);
483 | #endif
```

file_excerpt · crypto/nem.c · line 207 · SHA-256 ffe72c257d7484e9... · File c8858bed9e52cdac...

```
206 |                           const ed25519_secret_key private_key,
207 |                           ed25519_signature signature) {
208 |   if (private_key != NULL && signature != NULL) {
```

file_excerpt · crypto/nem.c · line 209 · SHA-256 19f8b44e8439c39d... · File c8858bed9e52cdac...

```
208 |   if (private_key != NULL && signature != NULL) {
209 |     ed25519_sign_keccak(ctx->buffer, ctx->offset, private_key, signature);
210 |   }
```

file_excerpt · crypto/nem.h · line 90 · SHA-256 89c1645bf199ca12... · File cdb89fd8d040261a...

```
 89 |                           const ed25519_secret_key private_key,
 90 |                           ed25519_signature signature);
 91 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12886 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry a role the engine determined and review agrees with, each checked against the cited file. Ed25519 at core/embed/sec/image/boot_header.c:81 and image.c is firmware and boot verification, with the public keys fixed in the bootloader. **It must not be called unmigrated,

> because this vendor has already started**: SLH-DSA appears in that same boot_header.c at line 30, there is a whole ML-DSA-44 module at core/embed/sec/mldsa44/, MCU attestation uses ML-DSA (core/embed/sec/mcu_attestation/), and the tree ships an X.509 certificate with an ML-DSA-44 public key (core/embed/sec/secret/unix/certs/T3W1.der). The inventory carries those seven rows as post-quantum. So the correct reading of this repository is a root of trust **mid-transition**, carrying both schemes, and the Ed25519 rows are what remains of the classical half rather than evidence that nothing has been done. Trezor is the most advanced repository in this portfolio on post-quantum readiness, and any report that presents it otherwise is wrong on a point its own engineers would see immediately. X25519 at core/embed/sec/tropic/tropic.c:383 plus crypto/noise_kk1.c and noise_xxpsk3.c is channel key exchange with the Tropic01 secure element over Noise — recordable now, decryptable later, and the standard answer is a hybrid exchange. ECDSA at crypto/bip32.c:40, crypto/secp256k1.c and crypto/zkp_bip340.c, and Schnorr (BIP-340) at crypto/zkp_bip340.c:139, are transaction signing. Ed25519 at crypto/bip32.c:477, crypto/nem.c and nem.h is transaction signing on the NEM path.

2 reference(s) to Schnorr (BIP-340) across 1 file(s), e.g. crypto/zkp_bip340.c:139. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Schnorr (BIP-340):transaction_signing
Data source	crypto/zkp_bip340.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crypto/zkp_bip340.c:139 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crypto/zkp_bip340.c · line 139 · SHA-256 0cf0a8147e0f6339… · File 05e714da983b70ec…

```
138 |   if (result == 0) {
139 |     if (secp256k1_schnorrsig_sign32(context_writable, signature_bytes, digest,
140 |                                     &keypair, auxiliary_data) != 1) {
```

file_excerpt · crypto/zkp_bip340.c · line 176 · SHA-256 713fd6fd76742347… · File 05e714da983b70ec…

```
175 |   if (result == 0) {
176 |     if (secp256k1_schnorrsig_verify(context_read_only, signature_bytes, digest,
177 |                                     32, &xonly_pubkey) != 1) {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12878 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: authentication and identity (certificates, tokens, device attestation) (2 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Correct and worth separating from transaction signing. The anchors are core/embed/sec/optiga/optiga_commands.c:1055 and optiga/unix/optiga.c — the Infineon OPTIGA secure element, which the device uses for device attestation and pairing rather than for spending. Short-lived authentication survives until a CRQC exists; a device identity that outlives the hardware does not, so this is a replace-at-next-hardware- revision item, not a firmware-update item.

2 reference(s) to ECDSA across 2 file(s), e.g. core/embed/sec/optiga/optiga_commands.c:1055. Role, judged from paths, function names and nearby code: authentication and identity (certificates, tokens, device attestation). Short-lived authentication stays safe until a CRQC exists; long-lived identities (device attestation, CA and SSH keys) need a plan earlier.

Affected asset	source-code:ECDSA:authentication
Data source	core/embed/sec/optiga/optiga_commands.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open core/embed/sec/optiga/optiga_commands.c:1055 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · core/embed/sec/optiga/optiga_commands.c · line 1055 · SHA-256 c6bd24690265c1b0... · File 9cfb47224dfdc540...

```
1054 |
1055 |   if (0 != ecdsa_sign_digest(&nist256p1, TA_PRIV_KEY, digest, &sop_cmd1[81],
1056 |                              NULL, NULL)) {
```

file_excerpt · core/embed/sec/optiga/unix/optiga.c · line 58 · SHA-256 fa6d6a3639322b2d... · File 8098dc92e54f1885...

```
  57 |   uint8_t raw_signature[ECDSA_RAW_SIGNATURE_SIZE] = {0};
  58 |   int ret = ecdsa_sign_digest(&nist256p1, DEVICE_PRIV_KEY, digest,
  59 |                               raw_signature, NULL, NULL);
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12881 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDH: role not determined (2 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Role resolved. crypto/bip32.c:702 is inside hdnode_get_shared_key: it calls curve25519_scalarmult(session_key + 1, node→private_key, peer_public_key + 1) and returns "ECDH is not supported" for any other curve. So this is ECDH between a BIP-32 node's private key and a peer public key, producing a session key for the identity and encryption features (SLIP-0017 style), not for spending. Harvest-now-decrypt-later applies to anything wrapped under such a session key, and because the node keys are derived from the device seed, a CRQC that recovers one node's private key recovers every session that node ever established. Priority: below transaction signing, above attestation.

3 reference(s) to ECDH across 2 file(s), e.g. crypto/bip32.c:702. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDH:general
Data source	crypto/bip32.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crypto/bip32.c:702 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crypto/bip32.c · line 702 · SHA-256 389be38816956858… · File 2429e49ef1cc89c5…

```
701 |     *result_size = 0;
702 |     return 1;  // ECDH is not supported
703 |   }
```

file_excerpt · crypto/zkp_ecdsa.c · line 33 · SHA-256 25cb88a1c594a8f8… · File bf9f86cfb6cd8826…

```
 32 | #include "vendor/secp256k1-zkp/include/secp256k1.h"
 33 | #include "vendor/secp256k1-zkp/include/secp256k1_ecdh.h"
 34 | #include "vendor/secp256k1-zkp/include/secp256k1_extrakeys.h"
```

file_excerpt · crypto/zkp_ecdsa.c · line 462 · SHA-256 3d362852669116be… · File bf9f86cfb6cd8826…

```
461 |
462 |   if (secp256k1_ecdh(context_writable, session_key, &public_key,
463 |                      private_key_bytes, plain_hash_function, NULL) != 1) {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12882 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: role not determined (3 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Role resolved. crypto/bip32.c:485 is inside hdnode_fill_public_key, which dispatches per curve: ed25519_publickey, ed25519_publickey_sha3, ed25519_publickey_keccak, curve25519_scalarmult_basepoint and the Cardano variant. The remaining anchors are crypto/ed25519-donna/ed25519.{c,h}, the vendored implementation. So these rows are the public-key derivation layer feeding the signing and ECDH paths that are inventoried separately, not an independent use. They belong in the inventory because the derivation itself has to change in a post-quantum scheme, but a reader should not count them as extra exposure on top of the signing rows.

6 reference(s) to X25519 across 3 file(s), e.g. crypto/bip32.c:485. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:X25519:general
Data source	crypto/bip32.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crypto/bip32.c:485 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crypto/bip32.c · line 485 · SHA-256 20d65e62022d8319… · File 2429e49ef1cc89c5…

```
484 |     } else if (node->curve == &curve25519_info) {
485 |       curve25519_scalarmult_basepoint(node->public_key + 1, node->private_key);
486 | #if USE_CARDANO
```

file_excerpt · crypto/bip32.c · line 696 · SHA-256 812181201a85f579… · File 2429e49ef1cc89c5…

```
695 |     }
696 |     curve25519_scalarmult(session_key + 1, node->private_key,
697 |                           peer_public_key + 1);
```

file_excerpt · crypto/ed25519-donna/ed25519.c · line 274 · SHA-256 822b93d7b53ae8cd… · File ef450fd2ea3296b3…

```
  273 | void
  274 | curve25519_scalarmult_basepoint(curve25519_key pk, const curve25519_key e) {
  275 | curve25519_key ec = {0};
```

file_excerpt · crypto/ed25519-donna/ed25519.c · line 303 · SHA-256 37ac0de6a16b455b… · File ef450fd2ea3296b3…

```
  302 | void
  303 | curve25519_scalarmult(curve25519_key mypublic, const curve25519_key secret, const
curve25519_key basepoint) {
  304 | curve25519_key e = {0};
```

file_excerpt · crypto/ed25519-donna/ed25519.h · line 29 · SHA-256 7441e37eaff8b03f… · File 9d9ceea846c11ecd…

```
   28 |
   29 | void curve25519_scalarmult(curve25519_key mypublic, const curve25519_key secret, const
curve25519_key basepoint);
   30 | void curve25519_scalarmult_basepoint(curve25519_key mypublic, const curve25519_key secret);
```

file_excerpt · crypto/ed25519-donna/ed25519.h · line 30 · SHA-256 64e240fedc50cea3… · File 9d9ceea846c11ecd…

```
   29 | void curve25519_scalarmult(curve25519_key mypublic, const curve25519_key secret, const
curve25519_key basepoint);
   30 | void curve25519_scalarmult_basepoint(curve25519_key mypublic, const curve25519_key secret);
   31 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12884 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (8 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Role resolved, and the anchor set is mixed, which matters for scoping. crypto/curves.c, crypto/ecdsa.c and crypto/ecdsa.h are the ECDSA implementation the device signs with — the correct and highest-priority inventory rows. But two of the eight files are not firmware: crypto/setup.py is the build script for the Python bindings, and crypto/tools/bip39bruteforce.c is a standalone

> demonstration utility. Neither ships on the device, so a migration plan should carry the three implementation files and drop those two. Flagging this rather than dismissing the whole finding: the rule is right, its file list is broader than the deployed surface.

50 reference(s) to ECDSA across 8 file(s), e.g. crypto/curves.c:25. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	crypto/curves.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crypto/curves.c:25 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crypto/curves.c · line 25 · SHA-256 af1973cfc17dac6a… · File 6117082c17ca7f8e…

```
24 |
25 | const char SECP256K1_NAME[] = "secp256k1";
26 | const char SECP256K1_DECRED_NAME[] = "secp256k1-decred";
```

file_excerpt · crypto/curves.c · line 26 · SHA-256 df8e4e49ae68afc4… · File 6117082c17ca7f8e…

```
25 | const char SECP256K1_NAME[] = "secp256k1";
26 | const char SECP256K1_DECRED_NAME[] = "secp256k1-decred";
27 | const char SECP256K1_GROESTL_NAME[] = "secp256k1-groestl";
```

file_excerpt · crypto/curves.c · line 27 · SHA-256 ed9fb77eedfc9250… · File 6117082c17ca7f8e…

```
26 | const char SECP256K1_DECRED_NAME[] = "secp256k1-decred";
27 | const char SECP256K1_GROESTL_NAME[] = "secp256k1-groestl";
28 | const char SECP256K1_SMART_NAME[] = "secp256k1-smart";
```

file_excerpt · crypto/ecdsa.c · line 38 · SHA-256 44a615006e3d1200… · File 76140e6cea8e4b04…

```
37 | #include "rfc6979.h"
38 | #include "secp256k1.h"
39 | #ifdef USE_SECP256K1_ZKP_ECDSA
```

file_excerpt · crypto/ecdsa.c · line 672 · SHA-256 e6ea0e71dbb0e64b… · File 76140e6cea8e4b04…

```
671 |   hasher_Raw(hasher_sign, msg, msg_len, hash);
672 |   int res = ecdsa_sign_digest(curve, priv_key, hash, sig, pby, is_canonical);
673 |   memzero(hash, sizeof(hash));
```

file_excerpt · crypto/ecdsa.c · line 983 · SHA-256 0809fafc1eba50e6… · File 76140e6cea8e4b04…

```
982 |   if (!curve) {
983 |     curve = &secp256k1;
984 |   }
```

file_excerpt · crypto/ecdsa.h · line 98 · SHA-256 9286f08eb43642fb... · File 206848978bfff6c3...

```
 97 |               int (*is_canonical)(uint8_t by, uint8_t sig[64]));
 98 | int ecdsa_sign_digest(const ecdsa_curve *curve, const uint8_t *priv_key,
 99 |                       const uint8_t *digest, uint8_t *sig, uint8_t *pby,
```

file_excerpt · crypto/ecdsa.h · line 130 · SHA-256 5ad92182d47e7850... · File 206848978bfff6c3...

```
129 |                 uint32_t msg_len);
130 | int ecdsa_verify_digest(const ecdsa_curve *curve, const uint8_t *pub_key,
131 |                         const uint8_t *sig, const uint8_t *digest);
```

file_excerpt · crypto/setup.py · line 18 · SHA-256 35d07f36d6767d0d... · File 7d35389b6dd2553f...

```
 17 |     "ripemd160",
 18 |     "secp256k1",
 19 |     "sha2",
```

file_excerpt · crypto/tools/bip39bruteforce.c · line 8 · SHA-256 de5f2fc1cd87824b... · File c9cda21c0539505e...

```
  7 | #include "ecdsa.h"
  8 | #include "secp256k1.h"
  9 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12885 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: role not determined (4 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Role resolved. All four anchors are the vendored ed25519-donna implementation, including its keccak and sha3 variants (crypto/ed25519-donna/ed25519-keccak.h, ed25519-sha3.h, ed25519.c, ed25519.h),

> which the NEM and Monero paths select through hdnode_fill_public_key. This is the Ed25519 implementation itself: the thing a migration replaces, reached by the transaction-signing rows already counted.

21 reference(s) to Ed25519 across 4 file(s), e.g. crypto/ed25519-donna/ed25519-keccak.h:10. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Ed25519:general
Data source	crypto/ed25519-donna/ed25519-keccak.h
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 19:34 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crypto/ed25519-donna/ed25519-keccak.h:10 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · crypto/ed25519-donna/ed25519-keccak.h · line 10 · SHA-256 9374afb8171aa7b3… · File b75b56fd205bb694…

```
    9 |
   10 | void ed25519_publickey_keccak(const ed25519_secret_key sk, ed25519_public_key pk);
   11 |
```

file_excerpt · crypto/ed25519-donna/ed25519-keccak.h · line 12 · SHA-256 f1505aedb366f82a… · File b75b56fd205bb694…

```
   11 |
   12 | int ed25519_sign_open_keccak(const unsigned char *m, size_t mlen, const ed25519_public_key
pk, const ed25519_signature RS);
   13 | void ed25519_sign_keccak(const unsigned char *m, size_t mlen, const ed25519_secret_key sk,
ed25519_signature RS);
```

file_excerpt · crypto/ed25519-donna/ed25519-keccak.h · line 13 · SHA-256 62dd018334b7d3d1… · File b75b56fd205bb694…

```
   12 | int ed25519_sign_open_keccak(const unsigned char *m, size_t mlen, const ed25519_public_key
pk, const ed25519_signature RS);
   13 | void ed25519_sign_keccak(const unsigned char *m, size_t mlen, const ed25519_secret_key sk,
ed25519_signature RS);
   14 |
```

file_excerpt · crypto/ed25519-donna/ed25519-sha3.h · line 10 · SHA-256 d6e35a345375ce78… · File 6e163e5a187b69d2…

```
    9 |
   10 | void ed25519_publickey_sha3(const ed25519_secret_key sk, ed25519_public_key pk);
   11 |
```

file_excerpt · crypto/ed25519-donna/ed25519-sha3.h · line 12 · SHA-256 c2198e821aab4c59… · File 6e163e5a187b69d2…

```
   11 |
   12 | int ed25519_sign_open_sha3(const unsigned char *m, size_t mlen, const ed25519_public_key pk,
const ed25519_signature RS);
   13 | void ed25519_sign_sha3(const unsigned char *m, size_t mlen, const ed25519_secret_key sk,
ed25519_signature RS);
```

file_excerpt · crypto/ed25519-donna/ed25519-sha3.h · line 13 · SHA-256 5acc5cdff3b6d75e... · File 6e163e5a187b69d2...

```
   12 | int ed25519_sign_open_sha3(const unsigned char *m, size_t mlen, const ed25519_public_key pk,
const ed25519_signature RS);
   13 | void ed25519_sign_sha3(const unsigned char *m, size_t mlen, const ed25519_secret_key sk,
ed25519_signature RS);
   14 |
```

file_excerpt · crypto/ed25519-donna/ed25519.c · line 47 · SHA-256 2375eee09341b31c... · File ef450fd2ea3296b3...

```
   46 | void
   47 | ED25519_FN(ed25519_publickey) (const ed25519_secret_key sk, ed25519_public_key pk) {
   48 | hash_512bits extsk = {0};
```

file_excerpt · crypto/ed25519-donna/ed25519.c · line 50 · SHA-256 e40eeeebef9f32d1... · File ef450fd2ea3296b3...

```
   49 | ed25519_extsk(extsk, sk);
   50 | ed25519_publickey_ext(extsk, pk);
   51 | memzero(&extsk, sizeof(extsk));
```

file_excerpt · crypto/ed25519-donna/ed25519.c · line 108 · SHA-256 5cf2afb8cd79ce2b... · File ef450fd2ea3296b3...

```
  107 | void
  108 | ED25519_FN(ed25519_sign_ext) (const unsigned char *m, size_t mlen, const ed25519_secret_key
secret_scalar, const ed25519_secret_key skext, ed25519_signature RS) {
  109 | ed25519_hash_context ctx;
```

file_excerpt · crypto/ed25519-donna/ed25519.h · line 12 · SHA-256 ab884f62531724e3... · File 9d9ceea846c11ecd...

```
   11 |
   12 | typedef unsigned char ed25519_signature[64];
   13 | typedef unsigned char ed25519_public_key[32];
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

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RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

Evidence

ID	Type	Data source	SHA-256
QSI-2026-12887	file_excerpt	common/protob/messages-benchmark.proto (line 2)	d58a0f922546e5589a782c4c431b203a948e0ba3367b16acc48b083757337b57
QSI-2026-12887	file_excerpt	common/protob/messages-bitcoin.proto (line 2)	d58a0f922546e5589a782c4c431b203a948e0ba3367b16acc48b083757337b57
QSI-2026-12887	file_excerpt	common/protob/messages-cardano.proto (line 2)	790478e3c9df02a9cc167174f39aa681bc17bc797f083b31cf5bc62de73da6d2
QSI-2026-12887	file_excerpt	common/protob/messages-eos.proto (line 2)	1bc71146f1570006a9a973ea956409f0e3a01cb63fa24947ee54d8bf842a73c5
QSI-2026-12887	file_excerpt	common/protob/messages-ethereum-eip712.proto (line 2)	1e20c49087fd2d8274025478654c869acc2208573fb52d553fd9ae6ad5b2e57b
QSI-2026-12887	file_excerpt	common/protob/messages-ethereum.proto (line 2)	2c715738dc6f29a41f531184d048ec35830475da4152cdbdc22a4ee88cc1327c
QSI-2026-12887	file_excerpt	common/protob/messages-monero.proto (line 2)	2a60fdbae9c4da472daabb0e85b95bb7592b561df57b5344c02f5d168a37c8d9
QSI-2026-12887	file_excerpt	common/protob/messages-nem.proto (line 2)	ae6d3c9d0ee3ae3e09e2007756938e4722cde7d467e71445ec9bdaa4af8d8e8a
QSI-2026-12887	file_excerpt	common/protob/messages-ripple.proto (line 2)	fc5096c71f30511dd5c0ea991b9a539f8aa06b6516b8317d4f738e6f164d30bb
QSI-2026-12887	file_excerpt	common/protob/messages-solana.proto (line 2)	bfc45618be53d3facd05de533c2d5bb4a77983730fe6ec3542fb7c63e67a6d17
QSI-2026-12887	file_excerpt	common/protob/messages-stellar.proto (line 2)	fad5a4b7a72a767742eb23629aef3942a53935e1c62cde7787ccad8cedb0e31a
QSI-2026-12887	file_excerpt	common/protob/messages-tezos.proto (line 2)	967d28a5f166e7d47d445277efcfecf0dc0f0596300340afd84afa0277ee829b

QSI-2026-12887	file_excerpt	common/protob/messages-tron.proto (line 2)	19b3678a04d39ff5ab88fc9368ffcc08a139179c086ed362f6b6a4b2391b46ea
QSI-2026-12877	file_excerpt	core/embed/sec/image/boot_header.c (line 81)	b2af0c1adee8c4fb6c493c712f1d917bb01ae6443327255e3d28e94b27eb762e
QSI-2026-12877	file_excerpt	core/embed/sec/image/image.c (line 190)	dfcc4a69e889974486837aa4be50157bff4a2e73e324925df33bc30df7fc03b3
QSI-2026-12877	file_excerpt	core/embed/sec/image/image.c (line 249)	708bc289abcbd866d09ef041c2d9869e54b53f060c69c50d2f9fd26d00edadd2
QSI-2026-12877	file_excerpt	core/embed/sec/image/image.c (line 411)	69cfc8f4b867ff5bea83d7fbadbbfe788411ba06e037fc81aa3ceca6b842628b
QSI-2026-12879	file_excerpt	core/embed/sec/tropic/tropic.c (line 383)	6a8b8eac9f12e02ddff658df23166595bf7d7db27950826444f652d9b3d16687
QSI-2026-12879	file_excerpt	crypto/noise_kk1.c (line 187)	92dfee834b4c368531df7ad9283f7d6639899197e7d98ba295c0e2ca0305c92f
QSI-2026-12879	file_excerpt	crypto/noise_kk1.c (line 199)	561606d6dbe7a1230e6670690b51b03eeb28f323fcc81250a5a75dd012373b11
QSI-2026-12879	file_excerpt	crypto/noise_kk1.c (line 205)	e17fd3f06c5cea712465187e110e0ea951f81d40d8ce732a92df067866302bc6
QSI-2026-12879	file_excerpt	crypto/noise_xxpsk3.c (line 138)	9648d44dace3fff12b561c2e3e511312ec7def54cb7275384d84c2778cdbb3df
QSI-2026-12879	file_excerpt	crypto/noise_xxpsk3.c (line 197)	4e128b42ab2d3d6fed69a9574f691e939ebc76321a124491053274cc9b5770eb
QSI-2026-12880	file_excerpt	crypto/bip32.c (line 40)	a9165bffdef5ae27e2ba39ddb3d8ba0c10eb9341570969040cd15e1128810354
QSI-2026-12880	file_excerpt	crypto/bip32.c (line 670)	81ebab5faffac9ea0878d776655e59b42e6187f392af97d4484cd8633ebb2b8b
QSI-2026-12880	file_excerpt	crypto/secp256k1.c (line 24)	aaefce0c635dd2025b251d5f2b3a6d7e5d4f3329ad49681589c1452ed3fcf659
QSI-2026-12880	file_excerpt	crypto/secp256k1.c (line 26)	3fcb800f4c815d9d84ae60c3e4f2b8a6a4de071b52686015e530cfcd254dcd05
QSI-2026-12880	file_excerpt	crypto/secp256k1.c (line 55)	7e89b542010777ba928e99a932d1a071fb882a321b72f98f8f748d3584428fe4
QSI-2026-12880	file_excerpt	crypto/secp256k1.h (line 32)	5257b883e95e417a1661c06e7caa265fe8170e9ff108435e83c354ca6b840dac

QSI-2026-12880	file_excerpt	crypto/zkp_bip340.c (line 30)	ef19a8a2b914c5235289dd3b646042c2401b62b2b6e4014c65666ba7ec2c07b6
QSI-2026-12880	file_excerpt	crypto/zkp_bip340.c (line 31)	860dd605d6fca510e08f9c516a9e05341047eef9ba000f45bcfc306db7ea4787
QSI-2026-12880	file_excerpt	crypto/zkp_bip340.c (line 32)	f0d785deb2e91ec837b7e79180e1ee76920a0e886315f256da82f00db6ae5a5f
QSI-2026-12883	file_excerpt	crypto/bip32.c (line 477)	c93b42f00e9370066521cf79c2798fde91c9b2822853563ab3afceb469dabc28
QSI-2026-12883	file_excerpt	crypto/bip32.c (line 479)	8ed678007d138957d9e205451ce947cd59213b5bbbb84462156ddd6ed04a303c
QSI-2026-12883	file_excerpt	crypto/bip32.c (line 482)	183902ffc93a30934c7802c54d27eec2bd33efac586d1eead5fbcac4656e92f8
QSI-2026-12883	file_excerpt	crypto/nem.c (line 207)	ffe72c257d7484e9296d04484e172589e186505be9b773b097a0e5e0aa0f47f2
QSI-2026-12883	file_excerpt	crypto/nem.c (line 209)	19f8b44e8439c39d9fd1783e94997f6dabee1f50815d190afaab2ec10e81c58e
QSI-2026-12883	file_excerpt	crypto/nem.h (line 90)	89c1645bf199ca12171f92e84bfa0d38b9fd2b1d2d4638a2e9aa3431e9ffdf38
QSI-2026-12886	file_excerpt	crypto/zkp_bip340.c (line 139)	0cf0a8147e0f6339f73b0bb98dcde6414ca590a73ef0343d41e17dcc3981242
QSI-2026-12886	file_excerpt	crypto/zkp_bip340.c (line 176)	713fd6fd7674234717caf272b5694b243a75dca531942f64b80fb738a30cdcb5
QSI-2026-12878	file_excerpt	core/embed/sec/optiga/optiga_commands.c (line 1055)	c6bd24690265c1b07db19daf28e91399ac13d6eb0cbc38c1205a7879e8cc5525
QSI-2026-12878	file_excerpt	core/embed/sec/optiga/unix/optiga.c (line 58)	fa6d6a3639322b2dfe9b568a69249366c4ac66f519d039a02e601df7e31f0449
QSI-2026-12881	file_excerpt	crypto/bip32.c (line 702)	389be3881695685895f920c4ceb0ad2dce6816c07e0808be71ddd0d9276f3d93
QSI-2026-12881	file_excerpt	crypto/zkp_ecdsa.c (line 33)	25cb88a1c594a8f8715ce5616f68f879732113346875c72386eae3a59ad78acb
QSI-2026-12881	file_excerpt	crypto/zkp_ecdsa.c (line 462)	3d362852669116be4ced1989d89e39cba9a5759a60bfd485b3a3957a9fe202b4
QSI-2026-12882	file_excerpt	crypto/bip32.c (line 485)	20d65e62022d83190469af099e68f304e380abf8949209c7259d85048d87b803

QSI-2026-12882	file_excerpt	crypto/bip32.c (line 696)	812181201a85f579d5d09a17ff9a49f0fa03fb8fcd9619daad1e577da9b7e1fe
QSI-2026-12882	file_excerpt	crypto/ed25519-donna/ed25519.c (line 274)	822b93d7b53ae8cde5036e6fca18fcafece86f3fbc88b6bc5cc54d1cb5797cf5
QSI-2026-12882	file_excerpt	crypto/ed25519-donna/ed25519.c (line 303)	37ac0de6a16b455ba9987423827a952280ef698f728ebf87d1abffd5eb988187
QSI-2026-12882	file_excerpt	crypto/ed25519-donna/ed25519.h (line 29)	7441e37eaff8b03f481182a533b231b8fa511fdd3274ae0e4e3e11d30856b7f4
QSI-2026-12882	file_excerpt	crypto/ed25519-donna/ed25519.h (line 30)	64e240fedc50cea38d023195f79cd62e4ad04135aa27e585b1d044f65f9eb20b
QSI-2026-12884	file_excerpt	crypto/curves.c (line 25)	af1973cfc17dac6a1989d2dd3431fe53093e5f4ec5cdcd8f9bcc9eca2182425
QSI-2026-12884	file_excerpt	crypto/curves.c (line 26)	df8e4e49ae68afc481ddc06f8dce5afbfe7451fba7984f22c7d5d57602f2471f
QSI-2026-12884	file_excerpt	crypto/curves.c (line 27)	ed9fb77eedfc925096d5cd8386f66c89e44624dc25a9b75b49b3c11861e85b25
QSI-2026-12884	file_excerpt	crypto/ecdsa.c (line 38)	44a615006e3d1200cc0d551f228b76e71bc6480255ec5e98e0c98d49cfb275fb
QSI-2026-12884	file_excerpt	crypto/ecdsa.c (line 672)	e6ea0e71dbb0e64b63b0e5cb5ceef206cf4767db7abb4f45a5b76f4a23a218dc
QSI-2026-12884	file_excerpt	crypto/ecdsa.c (line 983)	0809fafc1eba50e65dafc9e1fbeb55b5344f95d8c0a7474d1054332494da2a9c
QSI-2026-12884	file_excerpt	crypto/ecdsa.h (line 98)	9286f08eb43642fb5f95454a9d75f988673971caeab2b04687a958b20033c8e6
QSI-2026-12884	file_excerpt	crypto/ecdsa.h (line 130)	5ad92182d47e7850613b3230f1ccd1d44384b955791f31f7345f17c971ca1dc7
QSI-2026-12884	file_excerpt	crypto/setup.py (line 18)	35d07f36d6767d0d01fc07a769cd4a7df500e6c1d293d069fe8ef2f79547c1e6
QSI-2026-12884	file_excerpt	crypto/tools/bip39bruteforce.c (line 8)	de5f2fc1cd87824b1ddf275bb2273a6b7a26fe992025d825d1bbcb429b21a86c
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519-keccak.h (line 10)	9374afb8171aa7b315e6f8553b4065cfc7badf444f7da8747c0ebb0952478ab0
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519-keccak.h (line 12)	f1505aedb366f82a780a48e9d03c23168a3f8db8d473041211d4b29183f97ae4

QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519-keccak.h (line 13)	`62dd018334b7d3d19aade9cd71b6ec2e3489d20546ada5a4526524c1a452298b`
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519-sha3.h (line 10)	`d6e35a345375ce78aa3b055f726efa89bca19e06ebee6812d0860f85cb2beb0c`
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519-sha3.h (line 12)	`c2198e821aab4c59e07fcd1c2d91cc86f51e671156c91f0507e10651d290556c`
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519-sha3.h (line 13)	`5acc5cdff3b6d75ed5c1bef9fafbf2bd5939f58ea63f04710fd8141fd0cb9bc7`
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519.c (line 47)	`2375eee09341b31cd7358c1363832b24e0732231c1b4f60360bd51325ac231d2`
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519.c (line 50)	`e40eeeebef9f32d11972a9b630385dd03bbfd34fd827f55fa6e274bd440316f3`
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519.c (line 108)	`5cf2afb8cd79ce2bc6441aa80d3579ee0b34b2b90585209ca6135a4cdbee681c`
QSI-2026-12885	file_excerpt	crypto/ed25519-donna/ed25519.h (line 12)	`ab884f62531724e35b3e585259b46eeddf4ec6fb71606706c23b5537f02e91b2`

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 3 of 4 — hybrid protection in a critical path

- Hybrid — firmware and boot verification (root of trust): core/embed/sec/image/boot_header.c (Ed25519 + SLH-DSA)
- In use: SLH-DSA — core/embed/sec/image/boot_header.c
- In use: ML-DSA — core/embed/sec/mcu_attestation/inc/sec/mcu_attestation.h
- In use: ML-DSA — core/embed/sec/mcu_attestation/mcu_attestation.c
- In use: ML-DSA — core/embed/sec/secret/unix/certs/T3W1.der (DER) — T3W1 Trezor Safe 7
- In use: ML-DSA — core/embed/sec/secret_keys/inc/sec/secret_keys.h
- Present, not wired: ML-DSA — core/embed/sec/mldsa44/inc/sec/mldsa44.h
- Present, not wired: ML-DSA — core/embed/sec/mldsa44/mldsa44.c
- Classical only — channel key exchange (TLS, Noise, handshakes): X25519 (3 file(s))

Next step: Cover the remaining classical critical roles and plan when the classical half can be dropped.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
firmware and boot verification (root of trust)	Ed25519 (2)	QSI-2026-12877	Plan a hybrid firmware signature (the classical one plus ML-DSA or SLH-DSA, both required) and the path to ship the new keys in a bootloader update
channel key exchange (TLS, Noise, handshakes)	X25519 (3)	QSI-2026-12879	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)

transaction and message signing	ECDSA (17), Ed25519 (16), Schnorr (BIP-340) (14)	QSI-2026-12880, QSI-2026-12883, QSI-2026-12886, QSI-2026-12887	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
authentication and identity (certificates, tokens, device attestation)	ECDSA (2)	QSI-2026-12878	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
role not determined	ECDSA (8), Ed25519 (4), X25519 (3), ECDH (2)	QSI-2026-12881, QSI-2026-12882, QSI-2026-12884, QSI-2026-12885	Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
weak classically	4
reduced margin (Grover), approved	11
Grover: acceptable to NIST, below ECCG/ANSSI guidance	16
post-quantum	7
vulnerable (Shor)	74

Detailed findings: QSI-2026-12887, QSI-2026-12877, QSI-2026-12879, QSI-2026-12880, QSI-2026-12883, QSI-2026-12886, QSI-2026-12878, QSI-2026-12881, QSI-2026-12882, QSI-2026-12884, QSI-2026-12885

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit

Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDH (2), ECDSA (27), Ed25519 (29), Schnorr (BIP-340) (4), X25519 (12)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	BLAKE2 (5), ChaCha20-Poly1305 (2), HMAC (4), Keccak-256 (4), PBKDF2 (2), Poly1305 (3), RC4 (2), RIPEMD-160 (2), SHA-2 (2), SHA-3 (5), SLH-DSA (1)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	BLAKE2 (5), Keccak-256 (4), RIPEMD-160 (2), SHA-2 (2), SHA-3 (5)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.
Quantum lattice sieving Q1: quantum computation offline, classical access to the victim	included in standard parameters	ML-DSA (6)	No change for ML-KEM or ML-DSA at the standard parameter sets. A 2024 claim of a polynomial-time quantum LWE algorithm was withdrawn after a bug was found (see claims).

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.
- certificates — error (not checked): core/embed/sec/optiga/unix/certs/T2B1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- certificates — error (not checked): core/embed/sec/optiga/unix/certs/T3B1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- certificates — error (not checked): core/embed/sec/optiga/unix/certs/T3T1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- certificates — error (not checked): core/embed/sec/optiga/unix/certs/T3W1.der: not a PEM file and not a DER certificate either; it could not be parsed (ValueError)
- osv_advisories — not checked: no exactly pinned versions to query

Remediation plan

QSI-2026-12887 · P3 · 90 days — short-term risk reduction

Add ECDSA, Ed25519, Schnorr (BIP-340) usages at the wallet's transaction signing to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12877 · P3 · 90 days — short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12879 · P3 · 90 days — short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12880 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12883 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12886 · P3 · 90 days

short-term risk reduction

Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12878 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12881 · P3 · 90 days

short-term risk reduction

Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12882 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12884 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12885 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12887 · P4 · 90 days

root-cause fix

Plan migration of ECDSA, Ed25519, Schnorr (BIP-340) to ? (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.

Closure criteria	Re-audit shows ? or a hybrid scheme in use.

QSI-2026-12877 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-12879 · P4 · 90 days

root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-12880 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.

Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-12883 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-12886 · P4 · 90 days

root-cause fix

Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-12878 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.

Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-12881 · P4 · 90 days

root-cause fix

Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-12882 · P4 · 90 days

root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-12884 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.

Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-12885 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
None			

60 days

Priority	ID	Action	Type
None			

90 days

Priority	ID	Action	Type
P3	QSI-2026-12887	Add ECDSA, Ed25519, Schnorr (BIP-340) usages at the wallet's transaction signing to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12877	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12879	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12880	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12883	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12886	Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12878	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12881	Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12882	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction

P3	QSI-2026-12884	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12885	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-12887	Plan migration of ECDSA, Ed25519, Schnorr (BIP-340) to ? (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12877	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12879	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12880	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12883	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12886	Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12878	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12881	Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12882	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12884	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12885	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-12887	Re-run when a chain activates a post-quantum signature scheme or the wallet adds one.
QSI-2026-12877	Re-run the inventory after refactoring.
QSI-2026-12879	Re-run the inventory after refactoring.
QSI-2026-12880	Re-run the inventory after refactoring.
QSI-2026-12883	Re-run the inventory after refactoring.
QSI-2026-12886	Re-run the inventory after refactoring.
QSI-2026-12878	Re-run the inventory after refactoring.
QSI-2026-12881	Re-run the inventory after refactoring.
QSI-2026-12882	Re-run the inventory after refactoring.
QSI-2026-12884	Re-run the inventory after refactoring.
QSI-2026-12885	Re-run the inventory after refactoring.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.2.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.3.0
inventory_document	1.1.0
solidity	1.3.0
source_code	1.4.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.3.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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2. A. Chailloux, J. Loyer — Lattice sieving via quantum random walks (ASIACRYPT 2021; IACR ePrint 2021/570) — IACR ePrint. https://eprint.iacr.org/2021/570 (peer-reviewed article: 2^{0.2570d + o(d)}, previously 2^{0.2653d + o(d)} (Laarhoven))
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8. D. J. Bernstein — Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete? (2009) — cr.yp.to. https://cr.yp.to/hash/collisioncost-20090823.pdf (paper dated 2009-08-23)
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10. FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM) — NIST. https://csrc.nist.gov/pubs/fips/203/final (final (August 2024))
11. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
12. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
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140-validatable while one input comes from an approved scheme. Repeats the federal deadlines of OMB M-26-15: post-quantum key establishment by 2030-12-31 and signatures by 2031-12-31 for high value assets and high impact systems)

21. NIST — Post-Quantum Cryptography Standardization: Evaluation Criteria, Security (Call for Proposals, Section 4.A.5) — NIST. https://csrc.nist.gov/Projects/Post-Quantum-Cryptography/Post-Quantum-Cryptography-Standardization/Evaluation-Criteria/Security-(Evaluation-Criteria) (security categories: AES-128 key search 2^170/MAXDEPTH quantum gates, AES-192 2^233/MAXDEPTH, AES-256 2^298/MAXDEPTH; MAXDEPTH 2^40–2^96)
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Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
RC4		weak classically	broken	2	2	crypto/rc4.c; crypto/rc4.h
RIPEMD-160	RIPEMD-160	weak classically	legacy	2	2	crypto/ripemd160.c; crypto/ripemd160.h
ChaCha20-Poly1305	ChaCha20-Poly1305	reduced margin (Grover), approved	acceptable	2	2	crypto/chacha20poly1305/chacha20poly1305.c:13; crypto/chacha20poly1305/chacha20poly1305.h:13
HMAC		reduced margin (Grover), approved	acceptable	4	4	crypto/hmac.c; crypto/hmac.h; crypto/hmac_drbg.c
PBKDF2		reduced margin (Grover), approved	acceptable	2	2	crypto/pbkdf2.c; crypto/pbkdf2.h
Poly1305		reduced margin (Grover), approved	acceptable	3	3	crypto/chacha20poly1305/poly1305-donna-32.h; crypto/chacha20poly1305/poly1305-donna.c; crypto/chacha20poly1305/poly1305-donna.h
BLAKE2		Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	5	5	crypto/blake2_common.h; crypto/blake2b.c; crypto/blake2b.h
Keccak-256	Keccak-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	4	4	crypto/bip32.c:558; crypto/nem.c:135; crypto/sha3.c:397
SHA-2		Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	2	2	crypto/sha2.c; crypto/sha2.h
SHA-3		Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	5	5	crypto/ed25519-donna/ed25519-hash-custom-sha3.h; crypto/ed25519-donna/ed25519-sha3.c; crypto/ed25519-donna/ed25519-sha3.h

ML-DSA		post-quantum	acceptable	5	5	core/embed/sec/mcu_attestation/inc/sec/mcu_attestation.h:24; core/embed/sec/mcu_attestation/mcu_attestation.c:31; core/embed/sec/mldsa44/inc/sec/mldsa44.h:25
ML-DSA	ML-DSA-44	post-quantum	acceptable	1	1	core/embed/sec/secret/unix/certs/T3W1.der (DER) — T3W1 Trezor Safe 7
SLH-DSA		post-quantum	acceptable	1	1	core/embed/sec/image/boot_header.c:30
ECDH		vulnerable (Shor)	acceptable	2	2	crypto/bip32.c:702; crypto/zkp_ecdsa.c:33
ECDSA	secp256k1	vulnerable (Shor)	acceptable	17	17	common/protob/messages-benchmark.proto:2; common/protob/messages-bitcoin.proto:2; common/protob/messages-ethereum-eip712.proto:2
ECDSA	secp256k1, P-256	vulnerable (Shor)	acceptable	2	2	common/protob/messages-eos.proto:2; common/protob/messages-tezos.proto:2
ECDSA		vulnerable (Shor)	acceptable	8	8	core/embed/sec/optiga/optiga_commands.c:1055; core/embed/sec/optiga/unix/optiga.c:58; crypto/bip32.c:670
Ed25519	BIP32-Ed25519	vulnerable (Shor)	acceptable	1	1	common/protob/messages-cardano.proto:2
Ed25519	CLSAG ring signatures on the Ed25519 curve	vulnerable (Shor)	acceptable	1	1	common/protob/messages-monero.proto:2
Ed25519	Keccak-512 variant	vulnerable (Shor)	acceptable	1	1	common/protob/messages-nem.proto:2
Ed25519		vulnerable (Shor)	acceptable	4	4	common/protob/messages-ripple.proto:2; common/protob/messages-solana.proto:2; common/protob/messages-stellar.proto:2
Ed25519	Ed25519	vulnerable (Shor)	acceptable	22	22	core/embed/sec/image/boot_header.c:81; core/embed/sec/image/image.c:190; crypto/bip32.c:477
Schnorr (BIP-340)	secp256k1, Taproot	vulnerable (Shor)	acceptable	2	2	common/protob/messages-benchmark.proto:2; common/protob/messages-bitcoin.proto:2
Schnorr (BIP-340)		vulnerable (Shor)	acceptable	2	2	crypto/zkp_bip340.c:139; crypto/zkp_bip340.h

X25519	X25519	vulnerable (Shor)	acceptable	6	6	core/embed/sec/tropic/tropic.c:383; crypto/bip32.c:485; crypto/ed25519-donna/ed25519.c:274
X25519		vulnerable (Shor)	acceptable	6	6	crypto/ed25519-donna/curve25519-donna-32bit.c; crypto/ed25519-donna/curve25519-donna-32bit.h; crypto/ed25519-donna/curve25519-donna-helpers.c

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 90dd7412a16c9c32e0ab86bd7ea84f18277cb6f27e0cc88c4eafa5a87e5841cc

C. Input files

File	Size	SHA-256	Redactions
common/protob/check.py	1462	a1ffbf48b6690b8f01cbcfec154342846af9462ae5d47b23a3b2a2fbcd22a077	0
common/protob/graph.py	2082	1a60fccc6ea36a1339669ddb296f1b456b1bbe31419dc3b820e19e22d74cde28	0
common/protob/messages-benchmark.proto	923	8f6a6137c4158b2df1caf34b8d4fae8cf4acdb9968a37e5481bb657fae21b4c2	0
common/protob/messages-bitcoin.proto	28370	58e11f34625d55b9f62328a9b15b51843e02d34dd5da5d276c392c9d6db68b7f	0
common/protob/messages-ble.proto	741	67831e5b157e206dd1d3c2fc70b51b42160e8fd0b5f602a252dc2cd168c5dd0a	0
common/protob/messages-bootloader.proto	1116	9b0da32e12699c8f6ed114761f6a10cd06dfbc18cb649a50a238ac19962b2ed8	0
common/protob/messages-cardano.proto	18660	a6bec889024e85fe2f5bdc8f4730b975593443567b6cdaa7f189d6b1aa71a828	0
common/protob/messages-common.proto	7180	dbe669e87ef07922b0baf13b824337a3fc2b52b48f2c429970f3d76e59c80a35	0
common/protob/messages-crypto.proto	3191	835b80f7132972a7ca38848320cbc40223899c9950b17e1fcc11e3239f2d3b64	0
common/protob/messages-debug.proto	10196	6712380787106b9cc6fc9d31d36cf7d1b66b3eb838ca90d63629561e98a59c1e	0

common/protob/messages-definitions.proto	6388	9b4d80e3e6e9a801614a9255596a22f1e7006e5b393e3be8e1ab7349274f73e7	0
common/protob/messages-eos.proto	8913	1f78ce827ea132d06975f586fd492727d9b62436363488095c1b180ff4a09c32	0
common/protob/messages-ethereum-eip712.proto	3449	30fb535d393c3e61a0804b6298ef83beedc955eaf4da41ee071f16357dafd2f6	0
common/protob/messages-ethereum.proto	10198	33d2cdc65b0c0bf310be078947d753157417101dd8c638030766f101de078d34	0
common/protob/messages-evolu.proto	2725	cde2c188f13fab325586b5cfa84cfa8992157d6b82fa4284e7fb94e8f29c82a1	0
common/protob/messages-extapp.proto	3245	ea931433e9605db80b90afc41a1c2a3ab60d22da37d9134babfd82b1b8d2230c	0
common/protob/messages-management.proto	30329	1d1b306fc65adfaaa1b598e0948b70cc3259afac9ee0c7bd1196e3ca924509d1	0
common/protob/messages-monero.proto	15967	b5b5d9b971f5834db8b41901cf6e89fcfa491e6ac3e8bc26287465b8881ee4d2	0
common/protob/messages-nem.proto	9176	e6ecf0bbfe659e11be4866389d0e4c9f67b9e5fbb78e13f2460c3ecaf517eedb	0
common/protob/messages-nostr.proto	1809	778854486467aca41a9cf2678d309661017954bfbbdc8e42c6a9565a964727ef	0
common/protob/messages-ripple.proto	2896	c243dd611be61edf5f61ee968b23b63bfa6697cd043eca622f30e9bda81b7a35	0
common/protob/messages-solana.proto	3744	5132eea2e4101bf61d44e7c827d433aa82ed5b851d25e123c2244f456e969b0a	0
common/protob/messages-stellar.proto	25096	196c94ae6af8d1fc97c0f0ed30c78a2baa71ca4600da9dc4b2bdd31f15159773	0
common/protob/messages-telemetry.proto	845	43ba424447f061e44d3fde5e204dc032fd812e112f521e3331a7218a71967cf8	0
common/protob/messages-tezos.proto	5841	890233da5bd6398bb2de5037c4f4018269179b85eff85a46028f679fb707b934	0
common/protob/messages-thp.proto	9651	8e998a47a134e7a31cb8c26879dd714859ce0d4e68bb0c891c0baf0b680b3624	0
common/protob/messages-tron.proto	9885	f01e34d9ac5d45c1e208a78a818651e98f243ce1231c6f3b15119fad3068908b	0
common/protob/messages-webauthn.proto	1689	849f10aa7618f62032bfe1da7ebb5b9458af0441876c9a31c03fb32fc39c480c	0

common/protob/messages.proto	25205	a71b012dbb7d07f74c58cdc7a4afeb9616bb160ccd851e565156580a1d7f5687	0
common/protob/options.proto	3772	91cdd9232d1e08fef809b9814e33ab07a5a0e4885ea10ee71a1f9785106d27a7	0
core/embed/projects/bootloader/Cargo.toml	4129	6faa232eefde225a4e739da8b9a2a273e3cd76ca8e90cf9ab2b4bee96508e06d	0
core/embed/projects/bootloader/bootui.c	4687	6a551b35d4313e5f0c2fd9e1b33a3c1c0072231e2e3d51d696a612576414303c	0
core/embed/projects/bootloader/bootui.h	2668	7dc4f0217734cf6e580d5ae3779734041eab0cc9ea52c2f21caa7582fac31e98	0
core/embed/projects/bootloader/build.rs	2480	a275a9f3046d4314781186749d17615a7556dd429c050b4926e26e445325c593	0
core/embed/projects/bootloader/emulator.c	6765	d0393aaa563d021389ae61ec9315e5b022715c12173c403ba9689ef29dcc60a3	0
core/embed/projects/bootloader/emulator.h	849	77c4d5ef0cc6835fb444be4dd89f3fa3c4aaa92354575289bec9eb39c7830fb5	0
core/embed/projects/bootloader/fw_check.c	9048	369b3baa6960337efe794eaa7018ef32482a2aa836300bc109e91163597c1b04	0
core/embed/projects/bootloader/fw_check.h	4973	26f0183f8f4b2529d63c243b346ff5f92ef9d13c8711dfc7ceaef42d8d2c37ca	0
core/embed/projects/bootloader/header_pq.c	2590	72968e66a51497fd923272028f7f5440cf2358871b4f3d48742b9a958a15b313	0
core/embed/projects/bootloader/main.c	16005	91d266698a86858f4bd4acac9b6902ad11c0bb8923673506f22e2b2c96a7085c	0
core/embed/projects/bootloader/protob/pb/messages-debug.pb.c	579	30166dca7cf97a98e327c23aa185687e88a59d58f8eab6781268f45c2b6d8a25	0
core/embed/projects/bootloader/protob/pb/messages-debug.pb.h	13006	ef12062b1ff5e318aa88beeabc403c52cc3c9f95ec25cad709482288956841f8	0
core/embed/projects/bootloader/protob/pb/messages-debug.proto	4004	7bff5f5a52d9487be34df297354b9bebb0c5c5626f29b510388d94df7fd55fd9	0
core/embed/projects/bootloader/protob/pb/messages.pb.c	763	906fa5d22128433b3b478a76a4de3eef4f6bbdbfeecd7900db2d445f6e75e96f	0
core/embed/projects/bootloader/protob/pb/messages.pb.h	15305	aee15b545bdb3b9eef1cba9f82f264643c0c7f8ff4c4ab63f29a672e7c8647d1	0
core/embed/projects/bootloader/protob/pb/messages.proto	5529	3814ee49ce4767c6434a294451e77eef1671344f0305f5d9cbc47e595bb408ed	0

core/embed/projects/bootloader/protob/protob.c	7347	a909b4f76772c97553d58fd6beb77a7e30f06ca9bc10810dce2cf2e688f25ff7	0
core/embed/projects/bootloader/protob/protob.h	2308	65860166f5b7af173285b49d6963e6a44f75446e7c123c57d5338b9281ef746a	0
core/embed/projects/bootloader/protob/protob_common.h	3052	48cb5836996acfa321a9ca58e358da41190753886f5e1f2813fb6a633f6d0043	0
core/embed/projects/bootloader/protob/protob_debug.c	2920	c8973541cf462776b9b94d453ecc59a0b1fe60d720c3cebb6880cc514f95d11b	0
core/embed/projects/bootloader/protob/protob_debug.h	1365	a6b0c0b52b7f0c52105875b0933ce6b7a78db924fbbe5fe70f36fc588f6f41cc	0
core/embed/projects/bootloader/src/main.rs	125	9e6acc25712dac4982a96e90d6c740faebce1e78891df7b5863f6b00c77c754a	0
core/embed/projects/bootloader/ui_helpers.c	3135	380600c12f121c2f6433fbf745624488e7094552bea7985986b230c7eed2220d	0
core/embed/projects/bootloader/ui_helpers.h	951	7f7650a1737c6829a5502aa9c8fa03b8fa051f6d11b60cfa4a4a1f1fa63afda2	0
core/embed/projects/bootloader/version.h	414	3c5cf64b2af97f5a77b016d53aa7b8ea2e242976e3c2bf7f196fd5c5ef08f2b8	0
core/embed/projects/bootloader/version_check.c	3049	c2e62aaa755e802a113d7839cd407743330d2ecd94f50d676fa707e5353ddbf0	0
core/embed/projects/bootloader/version_check.h	1883	74e7c51d3e8737385e7da2e51f182b6e37419b2d560d9baa7b311dcf2e091319	0
core/embed/projects/bootloader/wire/codec_v1.c	7264	d03b7743f42e3f176fbfef8f0204fcb891c9678b5c4fec14fcd78aab95c3e30b	0
core/embed/projects/bootloader/wire/codec_v1.h	1811	b2c113dd0e7502aea3eb8fc01fae5d990ad0b74940ee0de190875dc358d8c0af	0
core/embed/projects/bootloader/wire/debug_iface_usb.c	2591	7ecedb924bc2a001bbd15f937fb20a849a6941480e0ae6a79266adf7daa9aec8	0
core/embed/projects/bootloader/wire/debug_iface_usb.h	909	0f1feaa29a0dc9b048d86e35891cc1829d6e57da0c6e06b9df71552c3308248b	0
core/embed/projects/bootloader/wire/wire_iface_ble.c	4548	0b1753453caa6eb671f9c0ab996ee46b48620c3ea00e08af5aa31aca61628610	0

core/embed/projects/bootloader/wire/wire_iface_ble.h	963	d091fec78cfb45f8d455f43e0a7770dc92c429818d04de7984bf351a8420a457	0
core/embed/projects/bootloader/wire/wire_iface_usb.c	2677	bd2014667fe0768fe1997f15bf64510312c8d254f48fc9fb9d743851d1d27bda	0
core/embed/projects/bootloader/wire/wire_iface_usb.h	931	4c2223b02ca7dc117c3329fb51f79641721ad6453ad92b655476f4b281899d24	0
core/embed/projects/bootloader/workflow/debuglink.c	5313	9ded890ae9c388785ff1a04cc94e2a37da2821c9ceccb8ec548dbc3dce40bde4	0
core/embed/projects/bootloader/workflow/debuglink.h	1030	033ef235c113f65d1dfd396dd0c6ac307cf3466dbf6731c43ac80006f7a94548	0
core/embed/projects/bootloader/workflow/wf_auto_update.c	1506	55fcaefcd28ef58441aec0c8dbc29a729c4a8bcfd85fe4810ad285232afb1c06	0
core/embed/projects/bootloader/workflow/wf_ble_pairing_request.c	5153	7c2a0197c1e2bcbc6dbd676ed4f9a68a6c8024be34277c901bc5b3897697d12b	0
core/embed/projects/bootloader/workflow/wf_bootloader.c	6709	c04c6b143de803a3f9f3e91e751ba4179a5ef20f65b214cb158b4f3e388578bd	0
core/embed/projects/bootloader/workflow/wf_empty_device.c	2528	c63aa59e1973af055fe81ab8d18f086e49d789805ba97e4d3833e218aa8a400a	0
core/embed/projects/bootloader/workflow/wf_firmware_update.c	16214	756c14cd1232b84b4df8eb5620b680e3519ec5a4dfe5c5f8f150cbba9b562d1e	0
core/embed/projects/bootloader/workflow/wf_get_features.c	1092	d9926653fd42f2785146cb508fafe7c265af7c71cdfa1446c2b5ca6218cd50a6	0
core/embed/projects/bootloader/workflow/wf_host_control.c	3593	60048eed5077390148f8a19ae1ef57edd1aa90c7c8d1ae5167fbbf66174f28a1	0
core/embed/projects/bootloader/workflow/wf_image_upload.c	12925	97384d25e91aa218ae444f5e78ee4dbc0f096b66c7488acffb118332178b8465	0
core/embed/projects/bootloader/workflow/wf_image_upload.h	7859	673f9ae576932eafa3981c47a47ab7663619f782943fd63b31e6a2ba0e5176e8	0
core/embed/projects/bootloader/workflow/wf_initialize.c	1085	f20614547336a484c9470b450bd91554a5d4eae9c8550c3071e39fd6731a7069	0

core/embed/projects/bootloader/workflow/wf_ping.c	1057	257907c1010f92d5fbff8b4225be8849dccc3c5e075c448c319bf39651bc9b7f	0
core/embed/projects/bootloader/workflow/wf_unlock_bootloader.c	1585	7b70feac57098fec83d6fdb2c14951e1835d17c869ec66377ef4d80c5013ffb7	0
core/embed/projects/bootloader/workflow/wf_wipe_device.c	3183	5dc051121baa1d70c14f0f6563e0ed694088e96e6ead48301266ff8bc2a62e5b	0
core/embed/projects/bootloader/workflow/workflow.h	2384	e407e9abd43800b10e5e3f4fbd7991133329656c9535049272c8122ec83420f4	0
core/embed/projects/bootloader/workflow/workflow_common.h	1263	bfa9440135ca5a4459795308c70e7132bf5e0d9ea3cbe5fc7987c3db0268a533	0
core/embed/sec/Cargo.toml	2374	01f264a9101775c0eb48517fec082bb1643e95f06886ea54d9907e2d5c2ac8b6	0
core/embed/sec/backup_ram/backup_ram_crc.c	1283	9a16d2b783c1197df005802bb4c109047da9bea104a8d7562534ca7b78575b70	0
core/embed/sec/backup_ram/backup_ram_crc.h	1386	d9edf2201a3c57d49f13c57ac7cfe49fe29448a5f5de2af2dce1223da09c21e8	0
core/embed/sec/backup_ram/build.rs	506	2d1283b95f503ff9fac8f4770c6a528955e9cf1d9d6d763597af8d5cd1cc2ef9	0
core/embed/sec/backup_ram/inc/sec/backup_ram.h	4424	86ae1f8e77c992f6ae3d4257b1ab9b45dd30b30bbb0107ee54a69146493611b8	0
core/embed/sec/backup_ram/stm32u5/backup_ram.c	14131	958852dc3fc1a6711fc781623167f51ae6cf893aa5a9589a9e74772c17c51599	0
core/embed/sec/backup_ram/unix/backup_ram.c	1442	e03af58cc7db205c71c0f5577bb02e12b64b0816aa2c38dfe1456d6f03dfe433	0
core/embed/sec/board_capabilities/build.rs	440	9b2b071dc5e8caa49d4477b183e674fd2099e6fe05fcdc22985be76ef7477d6f	0
core/embed/sec/board_capabilities/inc/sec/board_capabilities.h	2354	6972e1d6c6786bc2638f997289a0e118f7d8829353489512f47d6b353a773cb6	0
core/embed/sec/board_capabilities/stm32/board_capabilities.c	2366	68bd92926a40e379a7e28f934591bd30e9a1de0087b1e0a6605f039b6cb8a091	0

core/embed/sec/board_capabilities/unix/board_capabilities.c	1238	ac750ab207d3ac08ed5f34787e3cb4734ab656c5643cfd23d45ef3e3f69fe8e4	0
core/embed/sec/build.rs	3173	f8373796793e1f74f00bab4344f2571e1cbf8c212c79e09d08373584a05f0b6b	0
core/embed/sec/consumption_mask/build.rs	607	3832993d984fdd6482dec26222f25559141f7146440ff136f394ccf7f939fa55	0
core/embed/sec/consumption_mask/inc/sec/consumption_mask.h	951	51c6c87525c741b3d1dcf176063b668dfdfb6ce1a70af9b74727f24a07cbc381	0
core/embed/sec/consumption_mask/stm32f4/consumption_mask.c	4000	fb10fe34e200e9e5157fc591ed9d82547c1d593b116e004ea2e0ed90f0146053	0
core/embed/sec/consumption_mask/stm32u5/consumption_mask.c	5449	e71ad31a9754e0447230099ceb689b3973a8932e84521abaeca88d415d482560	0
core/embed/sec/consumption_mask/unix/consumption_mask.c	861	8370922fd2ed99c5a6aa468b770074a355b2d494a3abe39cf4019475e0fb5f48	0
core/embed/sec/fwutils/build.rs	179	df6e58b25c14babbb3f7b0e43903b471f21bbf19a6d5a3c5ce87572061bf7841	0
core/embed/sec/fwutils/fwutils.c	3732	1e9ba3553be64d1a9367e78ef35cb173c80a5d499d0039f0066047d17615c735	0
core/embed/sec/fwutils/inc/sec/fwutils.h	2151	75ba689a500f275efc5ee959fecc65f59fbb8b3c1586065c8355ef2afc84380e	0
core/embed/sec/hash_processor/build.rs	449	d2ff48c9a719cf3964cc956244f759f6ff1e8bb4a4435046ffc83782ecccb7fb	0
core/embed/sec/hash_processor/inc/sec/hash_processor.h	1819	f3d39b7a288559fc9b693c6e4977d3271f6f8e8b678573a880394659b161cc5c	0
core/embed/sec/hash_processor/stm32u5/hash_processor.c	5031	1c9eaecfe35d94793b57533e81e29b7787f997ea599af7afd81aaf53a71dff79	0
core/embed/sec/hw_revision/build.rs	456	1bf9a59aa0c0f0f41e8ce87c2e1c64ed6a60014e94c34d49015492cd9630567a	0
core/embed/sec/hw_revision/inc/sec/hw_revision.h	886	c3ec55542bb28dadba8b2c516c8a7f2fb23280bd348a4f097a8845d24989c894	0

core/embed/sec/hw_revision/stm32/hw_revision.c	3504	ae7401335382db8852a2e4a758e7bca580dc61c317061a684448b48d26da1c2c	0
core/embed/sec/hw_revision/unix/hw_revision.c	892	ea7adc245f9512046791e42e6107551137d0c869a2157b9093b49e178bf4d7cb	0
core/embed/sec/image/boot_header.c	8961	4506855d1b81cc19bb839d161266bed6404090dc9628c47de448fd17206b1df4	0
core/embed/sec/image/build.rs	802	d12f9a339b3d935fc9436d9fb1e7b46c2b48e195eccf1364f5f1ae82274bddc5	0
core/embed/sec/image/image.c	18805	3977fe12259f229c80c911094ff36164c6def2f15b20c36c6597edfac3997fef	0
core/embed/sec/image/inc/sec/boot_header.h	7371	ca6e16ae99fcf8c11ef8edfb7798b76a3597cf10e56d3f044f80d2dce255cf08	0
core/embed/sec/image/inc/sec/boot_image.h	2298	cbc400d0007a81fa57403075d53d2ad09cc80d3636299ca8fb9c6d0aa1583cdc	0
core/embed/sec/image/inc/sec/boot_ucb.h	2915	2b3ec18c71444c07bb42617ac64fce3e69e0df8cb05799a70f0d2e5e2318b6ef	0
core/embed/sec/image/inc/sec/image.h	6525	2654a3304e5ab4837b21c8dd7c8da6564abfdd879d3e203a615dd15d66870139	0
core/embed/sec/image/inc/sec/image_hash_conf.h	2230	71b096c8a9966115e7ee2aa0f3ebcbf1dd20bb1490e1528f7770eaf8292ab6f7	0
core/embed/sec/image/inc/sec/root_keys.h	3760	faa1c126b0259937d2d28dd42528dba2c97c84f23f74e41fa8cf745d918df268	0
core/embed/sec/image/stm32/boot_image.c	9817	d00f22374d6217e90f0dbed8b79f56771669e2d0db7e7985abb9152159511c93	0
core/embed/sec/image/stm32/boot_ucb.c	5523	d92657cf121e1c55702538bb48490d8e17d427756fc08f576ced87a31414821e	0
core/embed/sec/image/unix/boot_ucb.c	828	a42f7e49e37e82964facd48cc8d23e4e8255d55d74049aebb1f26c1fe74178df	0
core/embed/sec/iwdg/build.rs	414	1c7146464c3a4e0ba13ea8d29fea2ae8468644223bf7f522d50ac52dd083c55f	0
core/embed/sec/iwdg/inc/sec/iwdg.h	1288	dc19d91df2bdd576f9f8f042eb8796ca932fc7f90bd70b00c3e067753e34ed0e	0
core/embed/sec/iwdg/stm32/iwdg.c	1448	629e75f096e02af97e1bdc8c91b2d5133d7483eeeaeb2d3afc1a80d93857e2c6	0
core/embed/sec/iwdg/unix/iwdg.c	811	8fac0005b1a451692818f163cadb6405758872b55c396d9bed84aaa4ceaa5edc	0
core/embed/sec/mcu_attestation/build.rs	258	d57bf59e901394dbf2ba4d4e936b23d8279a4afa9f304b8c6af502fdeeccf23e	0

core/embed/sec/mcu_attestation/inc/sec/mcu_attestation.h	2357	873ee1bf6ae510a03ab9f28583e467d8b064a6bfd22a44eeba1d8c12deebe12f	0
core/embed/sec/mcu_attestation/mcu_attestation.c	2922	bdd4c58eb333727d7342865a7f21675cf6f6b7535fbe75cd40fd0fa20471e28d	0
core/embed/sec/mldsa44/build.rs	226	2f25187894fdf60d4acee296ef21e916a501ca5b11b5792820d19997d86db53a	0
core/embed/sec/mldsa44/inc/sec/mldsa44.h	1883	e1a692d1b32bfa3ee119f64b5816788e1913b1a882ae20e7e7d2fbd54db24464	0
core/embed/sec/mldsa44/mldsa44.c	1679	39d4e9751f5377b76009ce76b794b4fd5fa494569ee175873605f477369d108b	0
core/embed/sec/monoctr/build.rs	488	b57d07bdf2fb0c53f137c1cdff4d4553600eebfb0cfc81bac43c4d48f87ac44d	0
core/embed/sec/monoctr/inc/sec/monoctr.h	2111	78f1912acabf3b7d1997c6a541f75c852b45efe87c9b97b30c1dee332b9640e1	0
core/embed/sec/monoctr/stm32f4/monoctr.c	3434	c682eafe38b1e4e7cdc7b9e575de8ae1f6f3df9c579befc408e60147b6d4a7ff	0
core/embed/sec/monoctr/stm32u5/monoctr.c	3557	55717c1c6be1ec2a03f3dcbf1eaaa717d182af718bca0263cb3c48549d0ccf82	0
core/embed/sec/monoctr/unix/monoctr.c	3600	b7b154d11f4104a3a6d85c289803e83bb7ac71ee49930e342d2cd86022161c66	0
core/embed/sec/optiga/build.rs	920	e855d0f17e92e0c6d6a969431b2c7e138da7ee988cbc256fd8c27ccc37abc7d4	0
core/embed/sec/optiga/inc/optiga.h	912	0953ff667ee720a039f7e5dd5e2cfbd480dcfa9889e5081fe1f54b709a730c31	0
core/embed/sec/optiga/inc/sec/optiga.h	4208	02a723fa3941034aef998efcd920b76cac6a65ef93b120694c783401457702a0	0
core/embed/sec/optiga/inc/sec/optiga_commands.h	13186	6e8a7491fefe2d6533b17e7dcbff816a225d890fa9412deced552248b5173557	0
core/embed/sec/optiga/inc/sec/optiga_common.h	1695	97ee5ba6720c4bf877b3e2d2a0900e474dfc15b849b2ac49b7f655fb1a200ce0	0
core/embed/sec/optiga/inc/sec/optiga_hal.h	947	156c0de82bfea7951124674ec66813ca78f93aabb557e1c769fbd29973ed2060	0
core/embed/sec/optiga/inc/sec/optiga_init.h	2567	6ed55906c4a835ed72ad1e50b535e1a8eec1dfd70431a368201a4c720651c985	0

core/embed/sec/optiga/inc/sec/optiga_transport.h	1792	d710512d36f10ee51765760963695a1b84740110342e1a8e5df31319a0d22b24	0
core/embed/sec/optiga/optiga.c	43686	86953c6e9c6e270342c04d994c58eac57fd5d5ee233c3dce40421611c9b06dd8	0
core/embed/sec/optiga/optiga_commands.c	36849	9cfb47224dfdc5400ffccd64e0bdafc94598d870e43a06664163f15879e767d3	0
core/embed/sec/optiga/optiga_init.c	3578	1b2d1e9093b3dd759bd73c0d322f79bf293e48af69363ddc80059f84b821d24c	0
core/embed/sec/optiga/optiga_transport.c	26822	7f1cb8c2abcff0e93e5facfcb5870cc6491ce9ff4366cae35b18017202a61088	0
core/embed/sec/optiga/stm32/optiga_hal.c	2780	cfa0320e20151394bea8e25b509b10d9961a8134b4a78fbb01448e331c2bdcd1	0
core/embed/sec/optiga/unix/certs/T2B1.der	901	eeb363d0ae4658f0094309633221a6183bbebd810b2e0d049ff660d615c3cec3	0
core/embed/sec/optiga/unix/certs/T2B1.h	6419	9f1e10a773e4df55950999bb83ae7df31bf4dcaeff6bbaef4594bd27ba9ac4e5	0
core/embed/sec/optiga/unix/certs/T3B1.der	908	544579e03b3873704697ff3b9898f084fbed1eaaa7f80b845679ac9d1ad155bf	0
core/embed/sec/optiga/unix/certs/T3B1.h	6419	445bbe90652a89a5eeb0ff51c43221bd943a7a58d16a8b136159e9717dac61c1	0
core/embed/sec/optiga/unix/certs/T3T1.der	906	e8456966682be783d12a02143d7bf75a15d3a2aac410426e54f9f1b5adf8a9b8	0
core/embed/sec/optiga/unix/certs/T3T1.h	6419	04d5a15e77c8ff483a818ddfd70454c405c4912b1be9e46499ae0e89c6108863	0
core/embed/sec/optiga/unix/certs/T3W1.der	978	125146b2836658d22fb5303858eb2cd1662cbc16da245369f56e8b23a246e310	0
core/embed/sec/optiga/unix/certs/T3W1.h	6899	175695be34a0aa47ed7356a42b8e6875ce44595ea7d656e8da83de618ed0f8cc	0
core/embed/sec/optiga/unix/optiga.c	4931	8098dc92e54f18853be020b5ce9fca365ca33eafa694784bd8783b998ab768c8	0
core/embed/sec/optiga/unix/optiga_commands.c	857	7f6230fbe09ab1b9769d97cd588bc6c01b013a50f0fee4ff65b64bb2701db7c1	0
core/embed/sec/optiga/unix/optiga_hal.c	903	22efe05b14652fa9160e49120992e97483e015c19ec232015a877b583648baec	0
core/embed/sec/optiga/unix/optiga_mock.c	1823	ce5f984d391de93485ba0705ec47aeafaf45f4019689055fe28331f00afe4001	0
core/embed/sec/optiga/unix/optiga_transport.c	905	4c3f619cac65e74627ffe67cbcb228715ba92c5a179b4285b6bd82151d704f69	0
core/embed/sec/option_bytes/build.rs	492	c40f2aca2dc758d2505b22774d2dc6b396f6fe2d2fc42e7d3286c3a30dcf5149	0

core/embed/sec/option_bytes/inc/sec/option_bytes.h	1428	41d8df05143f7bc2090e7087c21cc85fdc0ae0f344fb432c047b499b37d0578a	0
core/embed/sec/option_bytes/stm32f4/option_bytes.c	5613	e5a9cea7638d839172bf64d4b261a03e1ad072eb4056b806438800d85f8f5316	0
core/embed/sec/option_bytes/stm32u5/option_bytes.c	8742	6bff0cb62703677c0de82a20c0924a3cb6059b93740abf02b39c8beae0017a18	0
core/embed/sec/random_delays/build.rs	495	1ca3687b6542f98797d93ef55c5f4c4c123650daacc260e85822c222a9d3f450	0
core/embed/sec/random_delays/inc/random_delays.h	919	49fe70229b9a9d1e424105337cae4c1e1a0824a735ff6003b2a995d9586f1ba2	0
core/embed/sec/random_delays/inc/sec/random_delays.h	2581	929bca79bfd664e6b8fb1e86916b5725846f7dbb9e5200de4a554f8a4dc34aef	0
core/embed/sec/random_delays/stm32/random_delays.c	5737	e0924b5c8f8ea5cb01f4ff4041a80939021ffc177b58932e6d7cf6ab9d5758c0	0
core/embed/sec/random_delays/unix/random_delays.c	844	94820067ebba247ca5752d69b80b46649666f68e8ed7166bca965908ffdf92bf	0
core/embed/sec/rng/build.rs	174	6a9f19933b06d841f3aafe11b4550bad639b996214cb9bdb4d7ce3a259d61fdf	0
core/embed/sec/rng/inc/sec/rng_strong.h	2191	2df4879d6c972e3bfd41fc86755ffc59b0e2325bcf71dd41f5465e6ad623a3bb	0
core/embed/sec/rng/rng_strong.c	2731	780f5365b0ba3d83afe6099307944042bd61b41d1976c97beacd923afb283694	0
core/embed/sec/rsod/build.rs	178	46e236b779048c6672182a6aec4daf3b537c5e228c6e31042575f2fae4d871ac	0
core/embed/sec/rsod/inc/sec/rsod_special.h	1711	00b8cc295a039f269d301fbb76841a98811c61a9e6bfcc6a5c056fcaad170829	0
core/embed/sec/rsod/rsod_special.c	2139	51108360fb408d79eddcc18c366c189cb2d383fcaf5693ee9060900371697b74	0
core/embed/sec/secret/build.rs	527	91d6aca75e97e8086102a0a4dcdafeaece7297b71cf9c8267d0ac693a83100d1	0
core/embed/sec/secret/inc/sec/secret.h	6136	2f811658255f16e3eadfdd1d79a6a20a39aa3f2fb1c30561c8af1ec410dc4e3f	0
core/embed/sec/secret/stm32f4/secret.c	5273	c90a1501e645ded27b95dc63eec5bda26607392403484ce894e4b3fbcc32d20a	0
core/embed/sec/secret/stm32u5/secret.c	19034	8e319207add55d057fdcc6ab0723ef6b13aa4568e50e9e40e72ef3014a88815d	0

core/embed/sec/secret/unix/certs/T3W1.der	4067	d6ab4e4371b9bb1de0399cc895da9c8750e05df69f3f72646941e128bb6e658d	0
core/embed/sec/secret/unix/certs/T3W1.h	27459	1fdf6d5b02bc9df85c9da4c2ec0783016bdb1c773e0bc668d1197b7118d98e88	0
core/embed/sec/secret/unix/secret.c	5732	6e72912f9e0e34e48d371f6e3c80b19a4c112bc42a610f67beaa7daa46feb928	0
core/embed/sec/secret_keys/build.rs	718	bef75c8d4a5aa6db90509d2a0ad197a21e03887062442ae99ba95f7c591c9460	0
core/embed/sec/secret_keys/inc/sec/secret_keys.h	2847	ae6d50f502a5502d465f1812b85d437563a9f5bf144daa7bf3bb839a90a9164f	0
core/embed/sec/secret_keys/secret_keys_common.c	4456	638c5e7674446634a43bf13eeddae4edd8b512e8010d1f22b5c55a970289856b	0
core/embed/sec/secret_keys/secret_keys_common.h	1847	6aac098add0fb4b7f231ee967e43201c233b1c4c0d68bac1dcdf009f7d9aeade	0
core/embed/sec/secret_keys/stm32f4/secret_keys.c	2183	cddf2e742228c0c4bc3c4aff7aca902d03caf0525c430d10356443c77359dc93	0
core/embed/sec/secret_keys/stm32u5/secret_keys.c	6713	7fffa6707930d72b6311d31ed09ffac7324ec62c0ba536b903a56cfc27b513fd	0
core/embed/sec/secret_keys/unix/secret_keys.c	3226	32745efdcabd75bb7aab533dd7144928e0f88a9ce5a755ad9fcea9a0dad73e67	0
core/embed/sec/secure_aes/build.rs	470	7198c9198353b848a3b62121e1ed97f2811e895bf7fc96a59a50c42781f80510	0
core/embed/sec/secure_aes/inc/sec/secure_aes.h	2097	2e88affb899ffcf6d2a5092fd2840d43803dd5d2a40ae78904bbea5f0e140e5b	0
core/embed/sec/secure_aes/inc/secure_aes.h	916	80ee83a0852f879f3e3034095793047010654495e4df71215c076083873f481b	0
core/embed/sec/secure_aes/stm32u5/secure_aes.c	7127	c024544b576d2f2be7834c7368295efa052b0e9e4c945b0c1c6c2dd45eff58e4	0
core/embed/sec/secure_aes/stm32u5/secure_aes_unpriv.c	5149	7c17aa64a955f5688e6a05babeca8aabc5de55e005091dc5a7f8fe6b7ccfcbbd	0
core/embed/sec/secure_aes/stm32u5/secure_aes_unpriv.h	912	3a97bdb9514522c3d4e15d3b20b6fb255d78102eb91a0ffad60951019ecac5e4	0

core/embed/sec/secure_aes/unix/secure_aes.c	832	43083eeaaf1f2861a5732dadf723fcf5a9724df7acb99c58f9388ba88144c244	0
core/embed/sec/src/lib.rs	658	9cdc881760e9754fa5b867e4f509dc14c4349ccd37913cc95a751fe360d80e69	0
core/embed/sec/storage/build.rs	1110	a20346f3dabdb08a088a13d3c9efbe7fda2d35c4e4f985d29d7ef7e576761b4f	0
core/embed/sec/storage/inc/norcow_config.h	1409	3e8815693e8c644f49596fed42ad06e09f459581caa50a83a0317a292a3200e1	0
core/embed/sec/storage/inc/sec/storage.h	1156	47224f5152ab718fdf672cfec365e7fda050371097bc29f1358c70c34a3f288d	0
core/embed/sec/storage/stm32f4/storage_salt.c	1967	af2067592fa9f342f4af6db31f9bd3b0e8fadd70b2578dd143c67812d1cae7bd	0
core/embed/sec/storage/stm32u5/storage_salt.c	2752	2df8cbe775631bce91c8017d7c6195860cd48a5a7fe81859281fe3dc2990ab58	0
core/embed/sec/storage/storage_salt.h	1451	f822bf3e2629653666e3c18246ced0cb6a5dc76a4f5d1189d74d2aebc4c649be	0
core/embed/sec/storage/storage_setup.c	1055	3ad894cbce0792c400936c479ac1c763db95d5f9f625ecedd0f981678c2ff498	0
core/embed/sec/storage/unix/storage_salt.c	1024	8818039a784d3467850674c2c0c15cc5aa7e8e213275b33976be1c1040182437	0
core/embed/sec/suspend/build.rs	418	bdadc93aeecf8459a46a308ebb9acaf18ae1dd5d824a67cb7b8f715aebf74b41	0
core/embed/sec/suspend/inc/sec/suspend_io.h	1456	dd9496271ba55340fc2df7ac6d4f6384a678d03f97c7e399a935fde0d4615041	0
core/embed/sec/suspend/stm32u5/suspend_io.c	1995	ea45b8f6a447575c8d7dad2f4b62a68a75937d57ac3f90f4af54113c9ec8fd5b	0
core/embed/sec/tamper/build.rs	430	f79aca7d78ba98d022b39ff7ddc8e3c21203f230f300736b3ed8d25d26a5322f	0
core/embed/sec/tamper/inc/sec/tamper.h	1263	2b88d5059803308db1c6f9ecbf9b53173746e37fbb8932bb176c4d5350f5fa96	0
core/embed/sec/tamper/stm32u5/tamper.c	8810	cebb07c63651dec077756fdad38d6d5c65e5ee99b77cb33deb0871949f780b03	0
core/embed/sec/tamper/unix/tamper.c	944	53b8b1a8ac82c33e2e0b4e5e3a04ccaa1e79021241cebad19dbfae8ec29e84da	0

core/embed/sec/telemetry/build.rs	448	d644e1f216527dc677e66f4d062a7c40a6adbd02b6d2d4b10f4633cc7f9564fa	0
core/embed/sec/telemetry/inc/sec/telemetry.h	2966	f4b8e43c54b4e1d43cc0e5e9f978c4362a0a6e5bdbe61b5a8e71265cbe1e0d85	0
core/embed/sec/telemetry/stm32u5/telemetry.c	3626	8c69f73e024b7a81f68041fc0c2c20e0f3edeb75ad6dc28abe6ab9a19b486ed7	0
core/embed/sec/telemetry/unix/telemetry.c	1200	906ec8c58badeae9418d99fae8ccf19d08fa69032e61ee4f10a91c4adead0ab9	0
core/embed/sec/time_estimate/build.rs	415	942afba329fc1eb90d1ac8a6ae766bf7f19afe1f485cec5956d9dd1d73d2b607	0
core/embed/sec/time_estimate/inc/sec/time_estimate.h	1011	6c714445e35a1468de49afc586eb2db0e7b124ca2331b31e28ff12cf04ec482b	0
core/embed/sec/time_estimate/inc/time_estimate.h	919	e767266afba79c54c19316fded82c0882e79b381aff2e4b859d6b2b88b3210ad	0
core/embed/sec/time_estimate/stm32/time_estimate.c	1414	655a8f5bd898141ea5bae7f88f3e95e3e3f70a35b6dd72e28646af330b65bc8c	0
core/embed/sec/time_estimate/unix/time_estimate.c	947	e3bd0549984801b1b55e60420623bb3e0a367e7d6ae8055a2b4154d7200b2310	0
core/embed/sec/tropic/build.rs	2550	d864604c158167d548473a34ef1cb44ecaa79a4f2eb330689a3f9f13807535be	0
core/embed/sec/tropic/config/tropic_configs.c	6144	fddf577fd4eae6b56506ede53fe63aa185d5587feb28f7e184135a97bb0fd16d	0
core/embed/sec/tropic/inc/sec/tropic.h	7641	3051664f352f8896f5d8d7d4b97f81bd88b97aaa2447a384c6e1af65242ebd2f	0
core/embed/sec/tropic/inc/sec/tropic_configs.h	1506	526aaffa7bdc782ac68df28980e15076833e5e1d71ec3a84535b16386f8ab5d7	0
core/embed/sec/tropic/stm32/tropic01.c	6902	0748452c297092a6e1c4878a2f115f83d57a2610eb6698803e1afe1f149bdc1c	0
core/embed/sec/tropic/tropic.c	63454	06c12cd103221953f5062cac17dcdac5ce30c90faf1c5765074b0059bff4acb2	0
core/embed/sec/tropic/unix/tropic01.c	1134	33e20cb495d39bd2132b9f5908958ff77640e67d0ab35c912cfea483f051b25c	0
core/embed/sec/tropic/unix/tropic_mock.c	1987	389b7503fef16d6e7e787747e5b548d6dbdd580ed8ab1578a4ab8934f62c06e2	0
core/embed/sec/trustzone/build.rs	496	3b44789edd289f74a9a2fdc070c47fb9c743bc6fcae42bc1c4e9b78e3268c745	0

core/embed/sec/trustzone/inc/sec/tz_init.h	853	963d7a8c19c0bd11a0b26aba72ba66bc73c43882149d1691d3403adf2f09fc83	0
core/embed/sec/trustzone/inc/sec/tz_periph.h	1596	978f7c95a712004fa8efe709e5d6f389ee9a99c498fed9184fa3f230a269dec5	0
core/embed/sec/trustzone/stm32u5/tz_init.c	15066	805220eab859d0b0a0c1a0c073d2222b18eb3742d7414799d00be3af55f22207	0
core/embed/sec/trustzone/stm32u5/tz_periph.c	1963	36c22839bdf5dade940d7e6d2b1c16e0b014e9b2d18aa1e03454a630b362a34f	0
core/embed/sec/trustzone/unix/tz_init.c	838	416a7977baf16d67c24fb260a284bdb99e9d46c6ba39b4ef033069344e15c31f	0
core/embed/sec/unit_properties/build.rs	425	24e070527e39579650b2123bfd2ee3b4ea659554db289489756249c162558ab4	0
core/embed/sec/unit_properties/inc/sec/unit_properties.h	4035	79a6188c32d7195f2f8476a684d9994e48341d838c8ddfd497b0c03f1c3a92b7	0
core/embed/sec/unit_properties/stm32/unit_properties.c	7224	eb93cf0996c2f07b66383b995a39c341d1ac95cffec4fd97bad17697ccda4dc0	0
core/embed/sec/unit_properties/unix/unit_properties.c	2596	7116803c69f0cccc1184262691d08f1ba4c4c488dc991b9627f5a3df472a79bc	0
crypto/address.c	3437	a114dd3fed92c28efc9ac86bb114359c64326075dd533e7cb7108873f6119462	0
crypto/address.h	1644	612ef87eff552dc85aa5e85dee2ecb43ff164b1867baacd09919eed665e8e914	0
crypto/aes/aes.h	8027	268e51fc6678770e88bb4a42e3047818d4d35239a49ba7c0c2a3d25edcda6677	0
crypto/aes/aes_modes.c	31255	08a8615b6b66c8965c026d7c5aaf9b56e3612f53d7f263fa3be956826cd31fcd	0
crypto/aes/aesccm.c	10364	beed09cbd374d23d19855991de1159e0cc79711ba8d3ac7a12599662aed9d715	0
crypto/aes/aesccm.h	1898	32eb3a758714d00b4cc3ba96444664d8e8396db38853484d9eb79d7c088961ba	0
crypto/aes/aescrypt.c	10385	bffc247b4231fdea460d6b587264b3bf60a1f2c8616797c605c2d4570894c7f6	0
crypto/aes/aesgcm.c	19524	99e1834ba9c2cb24ac680d0bf8517a0767d04694a4cfc46bc38482b5b2e187cd	0
crypto/aes/aesgcm.h	10723	dc03c25b255a671fba25ba14c28cc350207a15cb9bf904c277960e24aadf12c2	0
crypto/aes/aeskey.c	16156	06de2b55fbb4636f41882ef5e7a4898e58291783a41560cb24f7e7c8e6696b75	0
crypto/aes/aesopt.h	27763	eee8cfb072ea283e77cbd5492d2cfeb3c1bf288ccac29cd19af8e0178938c5d7	0
crypto/aes/aestab.c	15862	b56f0a709400d6a346ccef8aadc95a8c5f4da435dbdab6b9fe8083b2874688c3	0
crypto/aes/aestab.h	5162	54a0b84dfb358c0b1577f777c85cf9d90abea3c6f4ec5d587b7d230f123154d5	0
crypto/aes/brg_endian.h	1072	882c1a6b44d1a2825dfc45e1ff1521724a9c42d0630a50c5e03792c9b4611334	0

crypto/aes/gf128mul.c	14140	c8a16c99f928fcbda1e924d3f9d01944cb3cb31f6599fe4140103c3dd75f573d	0
crypto/aes/gf128mul.h	6831	bde5b69030831d35ad28849276db9e717ad91e613b7aec6cb9a72dd7f4a0fd7d	0
crypto/aes/gf_mul_lo.h	27314	697a0f03307c5c7e5164a4e36e9697066337eddf5464cbe4599f7d3029fe6aa8	0
crypto/aes/mode_hdr.h	11901	0d8e0a86a9b64f0f7b8c4a7540bb2037ad7fcae384eae2e113a7fa17aac09f72	0
crypto/base32.c	6224	9865ee24997300d0fa0c0c513c112dcea14725a2732b933b50375ccbd5b7e4b4	0
crypto/base32.h	1820	af06a1cef3f0e92ed35124527607a24c423513033223251b24c1f5a8ed62d7fe	0
crypto/base58.c	6905	9675f568be65fec72ddd6fdb849d8c30ad318cd3f4e1d44d522b27b8238497b7	0
crypto/base58.h	1915	f9e2ac2294b68ae67920e2050c776cfb34617dbce97a571307fbda879af37613	0
crypto/bignum.c	64369	e6051759e8a86f0d90f26ff2220b8c5c96ce51a1a8997015525c5848348d7fec	0
crypto/bignum.h	7306	1b92bc5469e786f1700510ae091592ca58fd8fa52e02c5622907950d76113eb3	0
crypto/bip32.c	25521	2429e49ef1cc89c504ae0815c890c5bac059606b775811c38e8bc1369be903f9	0
crypto/bip32.h	5628	d939c43c23ffa672e524ca8d6a57609d8593c665ad160aa5acd43158452ccaa6	0
crypto/bip39.c	9764	6baf82d7a522d2334c5d6777a9aa2452eeea1925891a14556dd15998ce546225	0
crypto/bip39.h	2513	3774828450685eae9f3c47af5c95d0a0062be6c35c73b61d194dcc9465d25c15	0
crypto/bip39_english.c	26292	7d594a31f97c7732ca5b88c1382ff83961607288c544cf3e339bbc5804eac021	0
crypto/blake256.c	6582	69d2f6baf60f9743f14ba9e76b80a7153b910b68a89a7e5ae653164fe2ad438d	0
crypto/blake256.h	2185	db53c134b10bdb781d99e69faa38ba204be21329e53c88a372aa1fff16848480	0
crypto/blake2_common.h	945	b35578aef4002e2d60ae670aafeacbbca08e68c9bcad31b53030d091bd0008c6	0
crypto/blake2b.c	9875	13a96853d7db5a69c15132204c7e7bb9d37633477feb244c3c72c64c2c2a913f	0
crypto/blake2b.h	1244	5ed610574dea1a47e0e352fc58b5b2a9ea4c051cc88833cd65732dd91105fdfe	0
crypto/blake2s.c	9549	b07dad1a25778fedef287824b1ed4b55a073fc96a28238231d5629ef51d2f5a6	0
crypto/blake2s.h	1244	1b63131f4ce1457c98e7733ab3e951765848ebb41bd59dae168df3b648451d4e	0
crypto/buffer.c	3695	9293e14aa8e257e3e3528fe5ccf44bf6d79fc3518edb6c7d8bf8b57f3541d7f3	0
crypto/buffer.h	2498	0cea3d5b7659011afc02c10424a923b5171d370fccd2dc99ce1fa28a54c15d6a	0
crypto/byte_order.h	1761	496a5a80346a0fb48ef3fbf53c0da2b69399e1de224d67962342657a9dfd77d1	0
crypto/cardano.c	10342	c28881b92bd59b7a9f8f3368675c2d5fbf53cb7fedcf523aad172c7d656eb61d	0
crypto/cardano.h	2189	d6e6bce678f522f0fe3d229cc1a4baa91e1bc6a3e9a05e4b93f81c9dcadfbe3a	0
crypto/cash_addr.c	6397	637e51700eb115326f3b039211731607d4bc4a2dc63ce953d46ca6eb315feec7	0
crypto/cash_addr.h	3533	92908d32e408da03300fa25867f65e02b5a770453295e6cda8cc983b61a84988	0
crypto/chacha20poly1305/chacha20poly1305.c	2505	a827ab1f975618f343ae2468cb8ff4b3aee307b84a69f66205fcca00a102489b	0

crypto/chacha20poly1305/chacha20poly1305.h	727	4e75c8d3a0044625d80b98b9500799da84fd387c28b72dcb0d90a3aec5465ee8	0
crypto/chacha20poly1305/chacha_merged.c	6500	410cfbaf6c0e6b986cc951f08d6991ef0a45b884dfc0dc6b21d82b8b8002c695	0
crypto/chacha20poly1305/ecrypt-config.h	5804	7ed8cab12d40e1f86fe2511d0c3c0332594bbb56eea0e9592e7d9c825c1876a9	0
crypto/chacha20poly1305/ecrypt-machine.h	1182	70dd1a45d677c3cfa62955fd1d3e125a31002f44f550e85a25364ae6cac159b9	0
crypto/chacha20poly1305/ecrypt-portable.h	6780	8a1f73bb364d19382983d2bc1a8ca568c4a5f7cc9dafa696ace5ebe954dbbcce	0
crypto/chacha20poly1305/ecrypt-sync.h	7851	2684a9c88e80749f88e94caea40b8ba57ddea17c5cfc06dc6c479d48d2b15c0f	0
crypto/chacha20poly1305/ecrypt-types.h	1175	7d0b98aad7734e79ad1a897d7c568d60547ece3d3f03e639fc274e8c9bcc1639	0
crypto/chacha20poly1305/poly1305-donna-32.h	6515	85bc63c1a468fe31a896c242be1cb7afe0cc9a262e5042911abc2cc5dbd1bcb6	0
crypto/chacha20poly1305/poly1305-donna.c	5075	f47eda9544fbbdace81f24914f988f2133f1614bafaabba0e7b2718309d81a9b	0
crypto/chacha20poly1305/poly1305-donna.h	658	db2194945f4aeb914e13efee982d365cd32cc16009febb6f68b51b6aaf17514e	0
crypto/chacha20poly1305/rfc7539.c	1770	57ddcc6e371ff5b9ce1221796c4a6bc7110259f3493a27c99b8c677e3a8c1f26	0
crypto/chacha20poly1305/rfc7539.h	351	699340a53d5d2c02ea95450966fddd210a0c06551d48c96895c9005a45c6e67e	0
crypto/chacha_drbg.c	4950	3a79708e2a3061b6908755577f7cd1a837bb2ff0edc56a6d9c8a9490f648c965	0
crypto/chacha_drbg.h	2746	a9315eb6531dda82f61958387002f42b02c17724474d92c20e93fb21f528a9e3	0
crypto/check_mem.h	1099	ec7d5a86c9745d80faa310ffbd07710267493b08e7f71bd3fb8e5ed9038fc6a0	0
crypto/consteq.c	1697	5bfe0f3b7a79c7d4e8cdd4d052273ae8d4f2e5180a07db0bcad07076220a07c7	0
crypto/consteq.h	1521	09b70ff94f5daf83a0bfa6bb76a1c110d19456f07b61c0982a16ce2e7b7e4fec	0
crypto/curves.c	1740	6117082c17ca7f8e0fee9b39218d4911bc111dcc0e0aa6c733c9c3c8b882e146	0
crypto/curves.h	1650	883ce4ab5e2417a94eaf1cd8817a10eeed6e55ea96a3fae65666586ba5c0ee9a	0

crypto/der.c	4447	7cd3558a5b529e73b53c5f35d19d0dc0d91680e41030c1f47cad0fbc0dfa78f2	0
crypto/der.h	2342	e054ea4c45ff29f9424c6642cbf22a7d94e9d7bcee4568749ebe45c7bca38363	0
crypto/ecdsa.c	44260	76140e6cea8e4b04b5e388f295bbd71f43471e8fdec252bb014a4ab9ee067478	0
crypto/ecdsa.h	7939	206848978bfff6c343e44eee2ce3b8ecda31a920b9f5618ef3eaaeb0d77175a5	0
crypto/ecdsa_internal.h	2421	7e6d403a2ac5ded03706456d464aa7de50db928f782931a67e3e65ad1220595e	0
crypto/ed25519-donna/curve25519-donna-32bit.c	26969	efaa64f1471b7e16a1a2b97f4f328501a80a177743d5700b18799fa338d30b21	0
crypto/ed25519-donna/curve25519-donna-32bit.h	2641	a4d39914f88b3daef07b81e26163c1b5c7161a85d6d339edf6dd830fa2c69d22	0
crypto/ed25519-donna/curve25519-donna-helpers.c	2414	aeed56aeef2e3d2136d0f4bcb2a7df8c82d608d21ad63f04d8ed6e0869a961cd	0
crypto/ed25519-donna/curve25519-donna-helpers.h	479	5f8a33382d9d4a53b7d88326a4104bd6e35e619676bcf7985146f2c25ffba4b4	0
crypto/ed25519-donna/curve25519-donna-scalarmult-base.c	2201	370a0c8b649ad61807a4220c9b961824dcb3b703f0af21135d04bb4d55930c7f	0
crypto/ed25519-donna/curve25519-donna-scalarmult-base.h	372	6b09a09393ce4a4d6cb0fde19db63785b5097d6521c6375fc664c3be8a5c6984	0
crypto/ed25519-donna/ed25519-donna-32bit-tables.c	11995	8345a35068d2fe7b39570a60644527a0265ce9cae75eaf70b98c2edb50595b26	0
crypto/ed25519-donna/ed25519-donna-32bit-tables.h	305	b8770e7c29cf8dabdda470537e56148a873519909b2de08f31a2dc8d9fe44f45	0
crypto/ed25519-donna/ed25519-donna-basepoint-table.c	124124	e35f8593c69d90a35a9694c6216eadc6ec79f660c984c0aa086d4b988a1a982d	0
crypto/ed25519-donna/ed25519-donna-basepoint-table.h	195	9edbe5f6537d72253acabe4a5c330827bd6802d17a57982837967b6f2ae4bc83	0
crypto/ed25519-donna/ed25519-donna-impl-base.c	22564	1724565c868af1d5af51d104b8ddd2ee6a6c85a6372107be2dd84d7c2c181bbe	0
crypto/ed25519-donna/ed25519-donna-impl-base.h	2793	1ea034e8b10f2f0474bbb64a0b71408ee0112126742e77b2042ab6d092f76970	0

crypto/ed25519-donna/ed25519-donna-portable.h	580	181fa91d854115b301633b6ebfb36ea28eef6528cefadc173592352f41ad516c	0
crypto/ed25519-donna/ed25519-donna.h	1218	51a561027071fcf644817152f43ee3ebbb7cf19bfdf102678058e3ab3638e84b	0
crypto/ed25519-donna/ed25519-hash-custom-keccak.h	785	04c8d8d1e9ed4c4cd63e88e24ba19f6eb56b04737c757730df7f69e032550ad3	0
crypto/ed25519-donna/ed25519-hash-custom-sha3.h	777	a0b36ebdb684c046933ff44a6293134274f75aee9a6cdd328af6caed55e8131e	0
crypto/ed25519-donna/ed25519-hash-custom.h	783	1bdce380cbe192dd220ee0786558d2c9d9a7814ebf2f58c46f87a0d0cabf4b42	0
crypto/ed25519-donna/ed25519-keccak.c	143	e80f13fd615ff1c457f077e08023f40845308d28a3016b485b8aad173499fc5b	0
crypto/ed25519-donna/ed25519-keccak.h	618	b75b56fd205bb694cb18c32e1054f68160a35f73a03fb9fc53db82b237713ad1	0
crypto/ed25519-donna/ed25519-sha3.c	137	036fc208ee7d4d2c75a700506b1c212bdd22fb075572f303544f7e8fbaab3e00	0
crypto/ed25519-donna/ed25519-sha3.h	604	6e163e5a187b69d2a53b12a5a22ec3502fcdb868c9d6d46f40f08f348dbf81ae	0
crypto/ed25519-donna/ed25519.c	7802	ef450fd2ea3296b3d9848e2b4a27d91807011103c95dca493339d6fc0319c55e	0
crypto/ed25519-donna/ed25519.h	1794	9d9ceea846c11ecd32ca4bc6e19b5169d7ecbc7b4dbabf62b335228df66b721f	0
crypto/ed25519-donna/modm-donna-32bit.c	23068	c4529c31b9f17c919b8d9c2fcddb05151c97de36c7c2dcf60dbb84c053e4a2d0	0
crypto/ed25519-donna/modm-donna-32bit.h	2703	6763fa27aadb7d26d83b825a0bd29a5ccc1662c6b42bd49ae794750b3bb4cd87	0
crypto/elligator2.c	5179	13e55ea9bf8db14d9bc6f0fba089696e5a77b0578e615d74624bef3222a58a32	0
crypto/elligator2.h	1344	ace382673072ce01ebdaf2fd262eb49bdc0747b14a8ac33c046cdfc1a40d0093	0
crypto/fault_handler.h	1228	a82155e446d326f0155a7ef6696b6e425e18cc4a59c50014e16e69ae36e2352d	0
crypto/fault_handler_noop.c	1352	314e9ecdde817e33f29e4139592e4803393fedc507fea043e324c00bffb35f19	0
crypto/groestl.c	29219	74c5b437ebe61325c78e508cfe23e9f7512188495ad8ba25f944bb20c8287b8f	0
crypto/groestl.h	3413	eb7ab0d6d707661f61411a3643fd5623fbe81f1fbc07304758c06c494417405b	0
crypto/groestl_internal.h	15138	360dd64ee6a7bb1411953ec37d8f9f5e3466ea3242571555f47e5c591225e1ce	0
crypto/gui/main.cpp	171	bf673474a73ac48682dedf65f57fdbee70bd5b6abb743e5566f58434a3ebe026	0
crypto/gui/mainwindow.cpp	2806	7f47cb8522f72158cfcf1bdf868fa9131e8ea648b9f8aa850dd23e63f94761d3	0

crypto/gui/mainwindow.h	412	aa4de2a09718dec1f4930220849602f244e9d3cd0f1c579294c8ba2918f5111d	0
crypto/hash_to_curve.c	11660	cc39ea65756d1f424f8f9643396b9515d34216e0eb3b41e9b1ffad7b31465c23	0
crypto/hash_to_curve.h	1584	6fce4d043d7808ad42616297d1f06100cc69387cf4402d27a047878ebb688fbb	0
crypto/hasher.c	4963	c6eab350c653e8769a33ba0cf0655b4cb1655021a69dccbac7f04065d51e434b	0
crypto/hasher.h	3410	4a423c86d326eda9bc1a55848e9eb7b5f3649edb2dc865261cb969769eacf55f	0
crypto/hmac.c	6389	ef44ae6fb924e2aa975e58789603e59399f9383887d3eda55e6920d23c627e24	0
crypto/hmac.h	2610	268e806610171b276d5e9bfb2fab84e825782b755eb62500c77ee812e50d4026	0
crypto/hmac_drbg.c	4708	9b82f03ecaf95cc1255a7e75e4ec4ecaba537bd7681f4a22a1fc6f00879ed201	0
crypto/hmac_drbg.h	1843	56cc90eded1de170f3d28b7c49ebe19072f824202d3c2e2f48a8dacf102690d2	0
crypto/memzero.c	1886	2f014e8d03420b42fb7b9df39f9150a33ac96d1c48a065f5a908a967561bca02	0
crypto/memzero.h	123	ffec031a236dde8df1c0b95cd0c3bd28ca046f6dc62faff36be8401ba82b214a	0
crypto/monero/base58.c	8545	7d5c45b0534c98f51918a7ca26bde2c7be2c8554e69019225fe278b024324bba	0
crypto/monero/base58.h	2297	eeb43561fa034917ab30921ab0d1c43ea49f4426ff2f719361b54fb28e20791c	0
crypto/monero/int-util.h	3223	43c09ffe3112b307a0725b67c88b983fb7ee5aef963d943158bafc48853e20bb	0
crypto/monero/monero.h	329	101b1cd67e8da011d098dfc18ee5027773ea38db17c09600c81ec4a402ffb8cc	0
crypto/monero/serialize.c	1073	5c10a6e62ce0527b7974406add5f97d2c20c883afb9aabf3b3116b9dabdfbda8	0
crypto/monero/serialize.h	361	f2a4135173f708918ed8bf802d8fbb2e10c6b3548cfe18d11d705925027350f5	0
crypto/monero/xmr.c	4383	28287d5ed78240a0ebaec6f3ecc941361fa2a1ddb144925ca7fa3422d95b2370	0
crypto/monero/xmr.h	2270	7de7ec3c2757158ef9734c0ddf706c44b99dd410ee6574ae42ecd0df3228ed71	0
crypto/nem.c	17425	c8858bed9e52cdac8606b46455d3af935a248941877b0360dc725e2044f74cc4	0
crypto/nem.h	6873	cdb89fd8d040261a129f20221e3c79814619857cf307266cbd6ed16d99e08bae	0
crypto/nist256p1.c	2625	76e4814899f47b5dea3380ff25da257b9b980d957d1fbbf1046e803a7be113f1	0
crypto/nist256p1.h	1360	1a8be02d270c3267eb5dc07465d305105c50c06498b19e7cd7a90b38e4776f67	0
crypto/noise_kk1.c	16163	2ddf35ff89fa3390d2c2bc86dd137a4426ed6cf889a88c104ed8a779c1120b9f	0
crypto/noise_kk1.h	5029	ccd8421270479065cddf8b425f6cee2f96fc455d1fead60ac3c96e47096f063e	0
crypto/noise_xxpsk3.c	30669	5550a2dcad0d68deb78bc1266fe4aaa6a4fe25e9c267593fa8840d5ece19c6ee	0
crypto/noise_xxpsk3.h	16382	928c7f768d721e9fc9b28793ced0d0218d86e7674e519c25bb0e3908519f6998	0
crypto/options.h	3023	762e3f5cf352740183330743d52e8f7aa977797038f97a3c0d8b88ff98074b1a	0
crypto/pbkdf2.c	6796	8482559322823c26ba3676eb281f229d4b0fc9d0cf22f9271629a1b8eef89736	0
crypto/pbkdf2.h	3084	8dfb7eea8ee9e0f99ff875a16d7fec538518227fbaa71fb0b9efc9a18b4e0483	0

crypto/rand.c	1615	a28436181f55ccccec8ea89f3598bba9342e0e8dad9284bac586dde0cf01a374	0
crypto/rand.h	1574	161ecc075038007a61fa7805b2d532bf36dabe521c3de480899eb4b8daa37c66	0
crypto/rand_insecure.c	2485	cac714a5c402542e3b85041d6cfa27f7f15459429f20264b21c0f06fa39a4f8e	0
crypto/rc4.c	1844	27d2f661c173af397dd954d14222291c0ff3016905dd618493c0868f1bf06ff8	0
crypto/rc4.h	1397	183634a8819215c6f1c0828ae37d91cb4672b0689c3bd6436a6b8b4168a099e6	0
crypto/rfc6979.c	2377	5e3f9afd203cdbfa411eff32aef0a3240f49664d3898e3c3027593ae001f501c	0
crypto/rfc6979.h	1680	876cd568de73effbfdf8b33b441d9fa4ea38456a4dd42b6312e44aca56a70c15	0
crypto/ripemd160.c	12630	819dbc7aa10c919feb7ede7aca303abd561379e5f816f680aab87a06f7181b7d	0
crypto/ripemd160.h	615	1c9cafb5cb0444ae815c138801252c6f0ffe3306e85180d952ff21be943b0305	0
crypto/script.c	2387	34c7128fcfe42f9fb88656893b61944898eadd7cc04981ce590a0bc328a8e0db	0
crypto/script.h	1346	797cd6d788b224ad800d1d7e1b5409643722aa86dc98f5cba3900e7e486c541f	0
crypto/secp256k1.c	3257	0a3762230f652a872f84a42b790b7c774be8a1a48564965d752f111ae5d1ce69	0
crypto/secp256k1.h	1501	ea553cc2e389005fd1b3041d3efbc169e57a03e1150f462f3f36e1858e612612	0
crypto/segwit_addr.c	7556	6f55007478e9988cc4fed000852f689bbf6869da08c8a0ff6ed59a91bbff4e94	0
crypto/segwit_addr.h	4295	5274a0701e4cac8d811c83f5c66873754af41fdbb89d414044d424acd55f3dad	0
crypto/setup.py	828	7d35389b6dd2553ff3f32e2c567d4c79fb86f59d058377a96a8854a054c91d59	0
crypto/sha2.c	39700	4e0a22b05988882a1f857084fc6c5a0731a115f5637e9946f65ccd23f361e24a	0
crypto/sha2.h	5035	acc5ba73f3ca604ebbaa34651d254eee7eaab808198ce60c42c015e723d9a656	0
crypto/sha3.c	11650	246f3f3f49b58803e654e72cc39d26451d7214fd47a6d594402304cc84aad3ba	0
crypto/sha3.h	3068	0e8f1617e73d948e60ce0c93598a1edea7c61f74b742d6caf408e2db0d9b9101	0
crypto/shamir.c	9937	f58a4b3a6f0c46f99a5ce3779b92ebd1064e1c46ca09a7be94cc3f14d3dad88d	0
crypto/shamir.h	3246	8d8a1286ab5cd315e0210862c2a61c354854c81ec412908b265d3a4e8f0a190c	0
crypto/slip39.c	4168	b486cf15acffa5f75d0c2c360bc476250c5860f6712b2d562cc7f88ec67b97b0	0
crypto/slip39.h	1589	dd177a5e49e17eb0e7def1be13e4b3e529c1fb890f83f5442e05cb0315bba738	0
crypto/slip39_english.c	13890	8fc3f9315190504ed455fc9da1cc88fdc2a4bef020c116edbb9b4d46a919c132	0
crypto/slip39_wordlist.h	31482	f8c5419144775a45bb6ec49a5a2e339b5b35006795065d4b914427a1dd8714b8	0
crypto/tls_prf.c	3255	a35878a414d0e4828067bcc8f5e6f3e0278f7d5da05287337e10072209463d84	0
crypto/tls_prf.h	1429	e2f86e529f96f208167b8118122da54756bff87fa06a6339afb607ecf2b3980a	0
crypto/tools/bip39bruteforce.c	2709	c9cda21c0539505e8d7503178bc33a1a3fbb0c610ec9b1660a21566bf9a0d7bf	0
crypto/tools/mktable.c	1954	d817e8d8badcf9ec5221550175dfb3f5ed15ad3f5a8fe46c0562e7b63f924dc7	0
crypto/tools/nem_test_vectors.rb	2644	99184a81184c8f4e522bbc5edf3d9fb25a83e4a8e7f2e193476b36e568f9c1d0	0

crypto/tools/xpubaddrgen.c	1290	7f5354bb6605cad3e42a8c649750025e33eba26140949bc8c08d518b4241c549	0
crypto/zkp_bip340.c	10496	05e714da983b70ec8a124dbc24e25d3481b34d08c59a75e50ce3b07679056977	0
crypto/zkp_bip340.h	1013	349bdf50fa07102e46ce6e03204caf808aa9e39a6a9918791b08a63dd330e6a2	0
crypto/zkp_context.c	3584	486a23bd0406f599915dff608709ff04d5d239888dc8a53483711287ee3e6341	0
crypto/zkp_context.h	471	5eed6687a0cb95b70dceb5519eb756bfd47235721fa93d25507a8d1b86a11501	0
crypto/zkp_ecdsa.c	15993	bf9f86cfb6cd8826db58eec012d766f88515633d7a382ce401291ab172970418	0
crypto/zkp_ecdsa.h	1789	96e11e8dedc258e63c6b60e475631075d67bbf0cd29b61bbcc8d4e0619d62227	0